Goodman-type lower bounds for graph homomorphism densities

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Abstract

For a finite graph H with $v(H)$ vertices, chromatic number $\chi(H)$, and chromatic polynomial P_H, define $\Phi_H(p) = (1-p)^{v(H)} P_H((1-p)^{-1})$. This polynomial interpolates the homomorphism densities of H in balanced complete multipartite graphons. We study the lower bound $t(H, W) \geq \Phi_H(p)$ on the homomorphism density of H in graphons W of edge density $p > 1 - 1/(\chi(H) - 1)$, which generalises Goodman's triangle bound. We prove the inequality for chordal graphs whose maximal cliques have equal size, for trivially perfect graphs, and for joins of a clique and an odd cycle. The proofs use entropy along clique trees and a convexity criterion for adding a universal vertex. We also prove that a nonbipartite graph satisfies the inequality at all sufficiently high edge densities if and only if its girth is odd. For a graph on n vertices whose girth is odd, the bound holds whenever $p > 1 - (72n^{12})^{-1}$. The proof combines Whitney's forest expansion with a spectral comparison of cycle densities. Finally, among the 117 nonbipartite graphs on at most six vertices with no isolated vertices or isolated edges, exactly 94 satisfy the bound. The main results are formally verified in Lean 4.

Keywords: graphon, homomorphism density, chromatic polynomial, Goodman's inequality, chordal graph, Turán graphon.

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1 Introduction

Goodman's inequality [5] gives the lower bound $p(2p-1)$ for the triangle homomorphism density of a graphon of edge density p. Equality holds for balanced complete k-partite graphons, whose edge densities are $1 - 1/k$. The exact minimum clique density at a given edge density was determined by Razborov [12] for triangles, by Nikiforov [11] for K_4, and by Reiher [14] for all cliques. In this paper we ask which graphs H satisfy the polynomial lower bound obtained by interpolating their densities in balanced complete multipartite graphons.

All graphs are finite and simple. For a graph H, write $V(H)$ and $E(H)$ for its vertex and edge sets, $v(H)$ and $e(H)$ for their cardinalities, $d_H(v)$ for the degree of a vertex v, and $\chi(H)$ for the chromatic number. Let K_s, P_s, and C_s denote the complete graph, path, and cycle on s vertices, respectively; K_0 is the empty graph. We use $\sqcup$ for disjoint union. A *graphon* is a symmetric measurable function $W\colon [0,1]^2 \to [0,1]$; see Lovász [9]. Its homomorphism densities are defined by

$$t(H, W) = \int_{[0,1]^{V(H)}} \prod_{uv \in E(H)} W(x_u, x_v) \prod_{v \in V(H)} \mathrm{d}x_v, \qquad p = t(K_2, W).$$

Let P_H be the chromatic polynomial of H and define

$$\Phi_H(p) = (1-p)^{v(H)} P_H\left(\frac{1}{1-p}\right). \tag{1.1}$$

The expression in (1.1), initially defined for $p < 1$, extends to a polynomial with $\Phi_H(1) = 1$, since P_H is monic of degree $v(H)$. For $\chi(H) \geq 2$, we say that H satisfies the *Goodman-type lower bound* if every graphon W satisfies

$$t(H, W) \geq \Phi_H(p) \qquad \text{whenever} \qquad p > \underline{p}_H := 1 - \frac{1}{\chi(H) - 1}. \tag{1.2}$$

For $H = K_3$, we have $\Phi_H(p) = p(2p - 1)$.

For an integer $k \geq 1$, the *Turán graphon* T_k is obtained by partitioning $[0, 1]$ into k sets of equal size and assigning value zero within each part and value one between distinct parts. It satisfies

$$t(K_2, T_k) = 1 - \frac{1}{k}, \qquad t(H, T_k) = \frac{P_H(k)}{k^{v(H)}} = \Phi_H(1 - 1/k). \tag{1.3}$$

Indeed, $t(H, T_k)$ is the fraction of maps $V(H) \to \{1, \dots, k\}$ that are proper colourings. We call the numbers $1 - 1/k$ *Turán densities*, and we say that T_k is *optimal* for H if it minimises $t(H, W)$ among graphons W of edge density $1 - 1/k$, that is, if

$$t(K_2, W) = 1 - \frac{1}{k} \quad \Longrightarrow \quad t(H, W) \geq t(H, T_k). \tag{1.4}$$

Thus (1.2) implies that T_k is optimal for H for every integer $k \geq \chi(H)$.

Im, Kim, and Yang [6, Section 8] introduced the polynomial (1.1) and the problem of classifying the graphs that satisfy (1.2). They also announced the chordal theorem for maximal cliques of equal size (Theorem 1.1), the results on clique joins in Section 3, and the obstruction for graphs of even girth (Proposition 4.17). We give complete proofs of these results and establish the further criteria and classifications stated below.

1.1 Chordal graphs

A graph is *chordal* if it has no induced cycle of length at least four. A *clique tree* of a graph is a tree whose vertices are its maximal cliques and in which the cliques containing any fixed vertex form a connected subtree. Such a tree exists for every connected chordal graph; see Gavril [3] and the account of clique trees by Blair and Peyton [1].

Theorem 1.1 (Chordal graphs with maximal cliques of equal size)**.** *Let H be a connected chordal graph whose maximal cliques all have the same size $k \geq 3$. Then H satisfies the Goodman-type lower bound.*

We deduce Theorem 1.1 from the following criterion, which also allows maximal cliques of different sizes.

Lemma 1.2 (Clique-tree criterion)**.** *Let H be a connected chordal graph of clique number $k \geq 3$. Let $C_1, \dots, C_m$ be its maximal cliques, and let T be a clique tree. Let $r_i = |C_i|$ and $s_e = |C_i \cap C_j|$ for $e = ij \in E(T)$, and define*

$$b_j = \#\{i : r_i \geq j\} - \#\{e \in E(T) : s_e \geq j\}, \qquad 1 \leq j \leq k. \tag{1.5}$$

If $b_1 \leq b_2 \leq \cdots \leq b_k$, then H satisfies the Goodman-type lower bound.

As we show in Section 2, b_j is the multiplicity of $j - 1$ as a root of P_H. The criterion therefore depends only on the chromatic polynomial. If every maximal clique has size k, then $b_j = m - \#\{e : s_e \geq j\}$ is nondecreasing, giving Theorem 1.1. The proof combines Lee's entropy construction along a tree [7, Theorem 2.6] with the Moon–Moser recurrence for clique densities [10].

1.2 Trivially perfect graphs and clique joins

A graph is *trivially perfect* if it has no induced P_4 or C_4; see Golumbic [4]. Equivalently, these graphs are generated from K_1 by disjoint union and addition of a universal vertex, a vertex adjacent to every existing vertex [2].

Theorem 1.3 (Trivially perfect graphs)**.** *Every trivially perfect graph with at least one edge satisfies the Goodman-type lower bound.*

Write $F \vee G$ for the join of F and G, obtained from their disjoint union by adding every edge between the two vertex sets. We prove Theorem 1.3 by showing that addition of a universal vertex preserves the bound under a convexity hypothesis. Specifically, Proposition 3.2 shows that if F satisfies the Goodman-type lower bound and Φ_F is nonnegative, nondecreasing, and convex on $[\underline{p}_F, 1]$ with $\Phi_F(\underline{p}_F) = 0$, then $K_1 \vee F$ satisfies the bound, and $\Phi_{K_1 \vee F}$ has the same properties on the corresponding interval. This also applies to odd cycles, using the following result.

Theorem 1.4 (Im, Kim, and Yang [6])**.** *For every odd integer $\ell \geq 3$ and every graphon W of edge density $p > 1/2$,*

$$t(C_\ell, W) \geq p^\ell - p(1-p)^{\ell-1} = \Phi_{C_\ell}(p). \tag{1.6}$$

The range $p > 2/3$ is proved in [6, Theorem 1.1], and the extension to $p > 1/2$ is announced in [6, Section 8]. The accompanying Lean development includes the proof on the full range; see Appendix A. The polynomial Φ_{C_ℓ} satisfies the required convexity conditions on $[1/2, 1]$, so repeated addition of universal vertices gives the following consequence.

Corollary 1.5 (Clique joins of odd cycles)**.** *For every integer $s \geq 0$ and every odd integer $\ell \geq 3$, the graph $K_s \vee C_\ell$ satisfies the Goodman-type lower bound.*

1.3 High densities and girth

We say that H *eventually satisfies* the Goodman-type lower bound if there exists $p_H < 1$ such that $t(H, W) \geq \Phi_H(p)$ holds for every graphon of edge density $p \geq p_H$. For a graph containing a cycle, the *girth* is the length of a shortest cycle.

Theorem 1.6 (Eventual validity and girth)**.** *Let H be a nonbipartite graph. The following are equivalent:*

(i) *For all sufficiently large integers k, T_k minimises $t(H, W)$ among graphons of edge density $1 - 1/k$.*

(ii) *The graph H eventually satisfies the Goodman-type lower bound.*

(iii) *The girth of H is odd.*

Moreover, if the girth of H is odd, then $t(H, W) \geq \Phi_H(p)$ for every graphon W of edge density $p \geq p_H$, where

$$p_H = 1 - \frac{1}{72v(H)^{12}}; \tag{1.7}$$

in particular, T_k is optimal for every integer $k > 72v(H)^{12}$.

The proof in Section 4 writes $W = 1 - U$ and expands $t(H, 1 - U)$, then groups the terms over forests using Whitney's broken-circuit theorem. We estimate this sum through a spectral comparison of cycle densities and bounds on the number of cycles in H. For even girth, the constant graphon violates the bound at every density sufficiently close to one.

1.4 Graphs on at most six vertices

Our last main result is a computer-assisted classification of small graphs. We identify graphs by their numbers in Read and Wilson's *An Atlas of Graphs* [13].

Theorem 1.7 (Classification on at most six vertices)**.** *There are* 117 *unlabelled nonbipartite graphs on at most six vertices with no isolated vertices and no isolated edges. Of these, the* 23 *graphs in Figure 2 violate the Goodman-type lower bound, and the other* 94 *satisfy it.*

Removing an isolated vertex or an isolated edge from H divides both sides of (1.2) by 1 or by p, respectively, and does not change $\chi(H)$. Hence a nonbipartite graph on at most six vertices satisfies the Goodman-type lower bound if and only if the graph obtained by deleting all its isolated vertices and isolated edges does.

The proof combines general density inequalities, estimates for individual graphs, and exact rational sum-of-squares certificates. Explicit graphons give the counterexamples. Each of the 117 cases has been formally verified in Lean. In Section 5, we also give density ranges on which the violating graphs satisfy the bound and show that, for exactly 100 of the 117 graphs, T_k is optimal for every integer $k \geq \chi(H)$.

1.5 Organisation

Section 2 proves the criterion for chordal graphs. Section 3 establishes the bound for trivially perfect graphs and clique joins of odd cycles. Section 4 proves the characterisation by girth and the quantitative threshold. Section 5 treats the classification on at most six vertices and optimality at Turán densities. Section 6 presents open problems, and Appendix A describes the formal verification.

2 A criterion for chordal graphs

We first express the chromatic polynomial through a clique tree. An entropy inequality then reduces the desired bound to a comparison of successive clique densities.

2.1 Clique-tree identities

We use the clique tree and notation of Lemma 1.2. For the standard structural properties, see [1]. Root the tree at C_1 and order its cliques so that each parent precedes its children. Set $S_1 = \varnothing$. For each C_i with parent $C_{\pi(i)}$, set $S_i = C_i \cap C_{\pi(i)}$. For every i, let $N_i = C_i \setminus S_i$. The defining property of a clique tree gives

$$C_i \cap \bigcup_{h<i} C_h = S_i.$$

The sets N_i partition $V(H)$, and $|N_i| = r_i - |S_i|$. Among the $\binom{r_i}{2}$ edges of C_i, precisely the $\binom{|S_i|}{2}$ edges within S_i have appeared in an earlier clique. Counting each vertex and edge in the first clique containing it therefore gives

$$\begin{aligned} v(H) &= \sum_{i=1}^{m} (r_i - |S_i|) = \sum_{i=1}^{m} r_i - \sum_{e \in E(T)} s_e, \\ e(H) &= \sum_{i=1}^{m} \left(\binom{r_i}{2} - \binom{|S_i|}{2} \right) = \sum_{i=1}^{m} \binom{r_i}{2} - \sum_{e \in E(T)} \binom{s_e}{2}. \end{aligned} \tag{2.1}$$

The last equality in each line uses $S_1 = \varnothing$ and the correspondence between each nonroot clique C_i and its edge to its parent in T.

Now fix an integer $x \geq k$ and colour the cliques in the same order. When C_i is reached, its already coloured vertices are exactly those in S_i, and their colours are distinct. The vertices in N_i must receive distinct colours chosen from the remaining $x - |S_i|$ colours. Thus each colouring of the preceding cliques has $\prod_{h=|S_i|}^{r_i-1}(x-h)$ extensions to C_i. Multiplying these numbers of choices gives

$$P_H(x) = \prod_{i=1}^{m} \prod_{h=|S_i|}^{r_i-1} (x-h) = \prod_{j=1}^{k} (x-j+1)^{b_j},$$

where the exponent of $x-j+1$ is $\#\{i : |S_i| < j \leq r_i\} = b_j$ by (1.5). Both sides are polynomials in x and agree for every integer $x \geq k$, so the identity holds for all x. In particular, b_j is the multiplicity of $j-1$ as a root of P_H. Each b_j is positive because H contains K_k. The colouring argument also gives $\chi(H) = k$. Since $v(H) = \sum_j b_j$, we obtain

$$\Phi_H(p) = \prod_{j=1}^{k} c_j(p)^{b_j}, \qquad c_j(p) = 1 - (j-1)(1-p). \tag{2.2}$$

2.2 An entropy inequality

We use entropy gluing as developed by Li and Szegedy [8], Szegedy [18], and Lee [7]. The construction below adapts Lee's Markov-tree theorem [7, Theorem 2.6] to graphons. When all maximal cliques have the same size, the resulting bound is the clique case of [7, Theorem 2.7]. The monotonicity of (b_j) extends the entropy comparison to cliques of different sizes.

Proposition 2.1. *Under the hypotheses of Lemma 1.2, every graphon W satisfies*

$$t(H,W) \prod_{j=1}^{k-1} t(K_j,W)^{b_{j+1}-b_j} \geq t(K_k,W)^{b_k}. \tag{2.3}$$

Proof. First assume $\varepsilon \leq W \leq 1$ for some $\varepsilon > 0$, which ensures all the entropy below are finite. For a random vector Z with coordinates in $[0,1]$ and probability density f, write $\mathcal{H}(Z) = -\int f \log f$ for its entropy. For a graph F, let p_F be the probability density on $[0,1]^{V(F)}$ obtained by normalising the product of its edge values. Evaluated at $Z = (Z_v)_{v \in V(F)}$, it is

$$\mathrm{p}_F\big((Z_v)_{v\in V(F)}\big) = \frac{1}{t(F,W)} \prod_{uv \in E(F)} W(Z_u, Z_v).$$

Applying Gibbs' inequality to f and p_F gives

$$\log t(F,W) \geq \mathcal{H}(Z) + \sum_{uv \in E(F)} \mathbb{E} \log W(Z_u, Z_v). \tag{2.4}$$

Equality holds when Z has density p_F.

Let $(X_1, \dots, X_k)$ have joint density

$$p(x_1, \dots, x_k) = \frac{1}{t(K_k,W)} \prod_{1 \leq a < b \leq k} W(x_a, x_b).$$

For $1 \leq j < k$, its first j coordinates have density

$$\frac{1}{t(K_k,W)} \int_{[0,1]^{k-j}} \prod_{1 \leq a < b \leq k} W(x_a, x_b)\, \mathrm{d}x_{j+1} \cdots \mathrm{d}x_k.$$

The joint density is invariant under coordinate permutations, so any j distinct coordinates have the same law as $(X_1, \dots, X_j)$.

We will construct a random vector $Y = (Y_v)_{v \in V(H)}$ along the rooted clique tree. On C_1, use the law of $(X_1, \dots, X_{r_1})$ defined above. For each subsequent clique C_i, order its vertices with those in $S_i = C_i \cap C_{\pi(i)}$ first. Conditional on their assigned values, sample the coordinates in N_i according to the conditional law of $(X_{|S_i|+1}, \dots, X_{r_i})$ given $(X_1, \dots, X_{|S_i|})$,

$$\begin{aligned} &p(x_{|S_i|+1}, \dots, x_{r_i} \mid x_1, \dots, x_{|S_i|}) \\ &\qquad \propto \int_{[0,1]^{k-r_i}} \prod_{1 \le a < b \le k} W(x_a, x_b)\, \mathrm{d}x_{r_i+1} \cdots \mathrm{d}x_k. \end{aligned}$$

The law already assigned to S_i agrees with this marginal, since S_i lies in the parent clique. Induction therefore shows that the coordinates on every C_i have the law of $(X_1, \dots, X_{r_i})$. By (1.5), with $b_{k+1} = 0$,

$$b_j - b_{j+1} = \#\{i : r_i = j\} - \#\{e \in E(T) : s_e = j\}.$$

The entropy chain rule gives

$$\begin{aligned} \mathcal{H}(Y) &= \sum_i \mathcal{H}(X_1, \dots, X_{r_i}) - \sum_{e \in E(T)} \mathcal{H}(X_1, \dots, X_{s_e}) \\ &= \sum_{j=1}^{k} \#\{i : r_i = j\}\, \mathcal{H}(X_1, \dots, X_j) - \sum_{j=1}^{k} \#\{e \in E(T) : s_e = j\}\, \mathcal{H}(X_1, \dots, X_j) \\ &= \sum_{j=1}^{k} \big(\#\{i : r_i = j\} - \#\{e \in E(T) : s_e = j\}\big) \mathcal{H}(X_1, \dots, X_j) \\ &= \sum_{j=1}^{k} (b_j - b_{j+1}) \mathcal{H}(X_1, \dots, X_j). \end{aligned}$$

Similarly, (2.1) gives

$$\begin{aligned} e(H) &= \sum_i \binom{r_i}{2} - \sum_{e \in E(T)} \binom{s_e}{2} \\ &= \sum_{j=1}^{k} \big(\#\{i : r_i = j\} - \#\{e \in E(T) : s_e = j\}\big) \binom{j}{2} \\ &= \sum_{j=1}^{k} (b_j - b_{j+1}) \binom{j}{2}. \end{aligned}$$

Every edge of H lies in a clique, so its two coordinates in Y have the law of (X_1, X_2). Applying (2.4) to Y and substituting these two identities, we obtain

$$\begin{aligned} \log t(H, W) &\ge \mathcal{H}(Y) + e(H) \mathbb{E} \log W(X_1, X_2) \\ &= \sum_{j=1}^{k} (b_j - b_{j+1}) \left(\mathcal{H}(X_1, \dots, X_j) + \binom{j}{2} \mathbb{E} \log W(X_1, X_2) \right). \end{aligned} \tag{2.5}$$

Applying (2.4) to $(X_1, \dots, X_j)$ and K_j gives

$$\mathcal{H}(X_1, \dots, X_j) + \binom{j}{2} \mathbb{E} \log W(X_1, X_2) \le \log t(K_j, W),$$

with equality for $j = k$ by the choice of the joint density. Since $b_j - b_{j+1} \le 0$ for $j < k$, substitution in (2.5) yields

$$\begin{aligned}
\log t(H, W) &\ge \sum_{j=1}^{k} (b_j - b_{j+1}) \left(\mathcal{H}(X_1, \dots, X_j) + \binom{j}{2} \mathbb{E} \log W(X_1, X_2) \right) \\
&= b_k \left(\mathcal{H}(X_1, \dots, X_k) + \binom{k}{2} \mathbb{E} \log W(X_1, X_2) \right) \\
&\quad - \sum_{j=1}^{k-1} (b_{j+1} - b_j) \left(\mathcal{H}(X_1, \dots, X_j) + \binom{j}{2} \mathbb{E} \log W(X_1, X_2) \right) \\
&= b_k \log t(K_k, W) \\
&\quad - \sum_{j=1}^{k-1} (b_{j+1} - b_j) \left(\mathcal{H}(X_1, \dots, X_j) + \binom{j}{2} \mathbb{E} \log W(X_1, X_2) \right) \\
&\ge b_k \log t(K_k, W) - \sum_{j=1}^{k-1} (b_{j+1} - b_j) \log t(K_j, W) \\
&= \log \left(\frac{t(K_k, W)^{b_k}}{\prod_{j=1}^{k-1} t(K_j, W)^{b_{j+1} - b_j}} \right).
\end{aligned}$$

Exponentiating proves (2.3).

For an arbitrary graphon, apply this result to $W_\varepsilon = \varepsilon + (1 - \varepsilon)W$ and let $\varepsilon \downarrow 0$. A telescoping product estimate gives $|t(F, W_\varepsilon) - t(F, W)| \le e(F)\varepsilon$ for every fixed finite graph F. All powers in (2.3) are nonnegative integers, so the inequality passes to the limit. □

2.3 Adjacent clique densities

The following is the graphon form of the classical clique recurrence of Moon and Moser [10, inequality (10)].

Lemma 2.2 (Moon–Moser recurrence)**.** *For every $j \ge 2$ and every graphon,*

$$j t(K_j, W)^2 \le t(K_{j-1}, W) t(K_j, W) + (j - 1) t(K_{j-1}, W) t(K_{j+1}, W). \tag{2.6}$$

Lemma 2.3. *If $p > 1 - 1/(k - 1)$, then $t(K_j, W) > 0$ for $1 \le j \le k$ and*

$$\frac{t(K_j, W)}{t(K_{j-1}, W)} \ge 1 - (j - 1)(1 - p), \qquad 2 \le j \le k.$$

Proof. We have $t(K_1, W) = 1$ and $t(K_2, W) = p > 0$, giving the bound for $j = 2$. Suppose $2 \le j < k$ and both inequalities hold up to j. Then (2.6) gives

$$\begin{aligned}
t(K_{j+1}, W) &\ge \frac{t(K_j, W)}{j - 1} \left(j \frac{t(K_j, W)}{t(K_{j-1}, W)} - 1 \right) \\
&\ge \frac{t(K_j, W)}{j - 1} \big(j(1 - (j - 1)(1 - p)) - 1 \big) \\
&= \big(1 - j(1 - p)\big) t(K_j, W) > 0.
\end{aligned}$$

Here $1 - j(1 - p) > 0$ because $j \le k - 1$ and $p > 1 - 1/(k - 1)$. This proves the lemma by induction. □

Proof of Lemma 1.2. Since $t(K_0, W) = t(K_1, W) = 1$, Proposition 2.1 and Lemma 2.3 give, for $p > 1 - 1/(k-1)$,

$$t(H, W) \geq \frac{t(K_k, W)^{b_k}}{\prod_{j<k} t(K_j, W)^{b_{j+1}-b_j}} = \prod_{j=1}^{k} \left(\frac{t(K_j, W)}{t(K_{j-1}, W)} \right)^{b_j} \geq \prod_{j=1}^{k} c_j(p)^{b_j} = \Phi_H(p).$$

□

The graph H in Figure 1 has maximal cliques $C_1 = \{1, 2, 3, 5\}$, $C_2 = \{1, 2, 3, 4\}$, and $C_3 = \{4, 6, 7\}$. The path C_1-C_2-C_3 is a clique tree, with clique sizes $4, 4, 3$ and intersections of sizes $3, 1$. Thus $(b_1, b_2, b_3, b_4) = (1, 2, 2, 2)$, and Lemma 1.2 proves the Goodman-type lower bound for H.

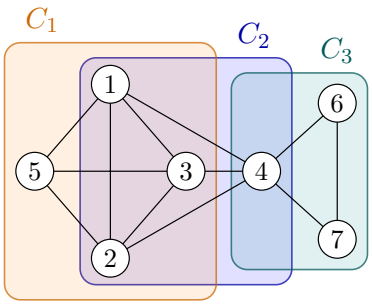


Figure 1: The chordal graph H and its maximal cliques C_1, C_2, C_3.

3 Universal vertices

We prove that adding a universal vertex preserves the bound when Φ_F is nonnegative, nondecreasing, and convex on the relevant interval. The proof begins with a weighted inequality for triangle densities.

3.1 A weighted triangle inequality

For a graphon W, let $d_W(x) = \int W(x, y)\, \mathrm{d}y$ be its degree function. Set

$$\tau_W(x) = \int W(x, y) W(x, z) W(y, z)\, \mathrm{d}y\, \mathrm{d}z,$$
$$M_j(W) = \int d_W(x)^j\, \mathrm{d}x \quad (j \geq 0).$$

Thus $\tau_W(x)$ is the triangle density in W with one vertex fixed at x. For a graphon V, let $(T_V f)(x) = \int V(x, y) f(y)\, \mathrm{d}y$ denote its integral operator. We use the inner product $\langle f, g \rangle = \int f(x) g(x)\, \mathrm{d}x$ on real $L^2[0, 1]$.

Lemma 3.1. *If $p = \int W(x, y)\, \mathrm{d}x\, \mathrm{d}y > 0$, then, for every integer $h \geq 0$,*

$$\int d_W(x)^h \tau_W(x)\, \mathrm{d}x \geq \left(2 - \frac{1}{p}\right) M_{h+2}(W). \tag{3.1}$$

Proof. Applying $ab \geq a + b - 1$ to $a = W(x, y)$ and $b = W(x, z)$, and then using the symmetry

of W, we obtain

$$\begin{aligned}\tau_W(x) &= \int W(x,y)W(x,z)W(y,z)\,\mathrm{d}y\,\mathrm{d}z \\ &\geq \int (W(x,y)+W(x,z)-1)W(y,z)\,\mathrm{d}y\,\mathrm{d}z \\ &= \int W(x,y)d_W(y)\,\mathrm{d}y + \int W(x,z)d_W(z)\,\mathrm{d}z - p \\ &= 2(T_W d_W)(x) - p.\end{aligned}$$

Multiplying by $d_W(x)^h \geq 0$ and integrating in x gives

$$\begin{aligned}\int d_W(x)^h \tau_W(x)\,\mathrm{d}x &\geq 2\int d_W(x)^h (T_W d_W)(x)\,\mathrm{d}x - p\int d_W(x)^h\,\mathrm{d}x \\ &= 2\langle d_W^h, T_W d_W\rangle - pM_h(W).\end{aligned}$$

Set $U = 1 - W$ and $q = 1 - p$, so that $d_U = 1 - d_W$. Since $T_W d_W = d_W - q + T_U d_U$,

$$\begin{aligned}\langle d_W^h, T_W d_W\rangle &= \int d_W(x)^h (T_W d_W)(x)\,\mathrm{d}x \\ &= \int d_W(x)^h (d_W(x) - q + (T_U d_U)(x))\,\mathrm{d}x \\ &= \int d_W(x)^{h+1}\,\mathrm{d}x - q\int d_W(x)^h\,\mathrm{d}x + \int d_W(x)^h (T_U d_U)(x)\,\mathrm{d}x \\ &= M_{h+1}(W) - qM_h(W) + \langle d_W^h, T_U d_U\rangle.\end{aligned}$$

Since $d_U(x) = \int U(x,y)\,\mathrm{d}y$, symmetry of U gives

$$\begin{aligned}&2\left(\langle d_W^h, T_U d_U\rangle - \int d_W(x)^h d_U(x)^2\,\mathrm{d}x\right) \\ &\quad= 2\int U(x,y) d_W(x)^h d_U(y)\,\mathrm{d}x\,\mathrm{d}y - 2\int U(x,y) d_W(x)^h d_U(x)\,\mathrm{d}x\,\mathrm{d}y \\ &\quad= 2\int U(x,y) d_W(x)^h (d_U(y) - d_U(x))\,\mathrm{d}x\,\mathrm{d}y \\ &\quad= \int U(x,y) d_W(x)^h (d_U(y) - d_U(x))\,\mathrm{d}x\,\mathrm{d}y \\ &\qquad+ \int U(x,y) d_W(y)^h (d_U(x) - d_U(y))\,\mathrm{d}x\,\mathrm{d}y \\ &\quad= \int U(x,y)(d_U(y) - d_U(x))(d_W(x)^h - d_W(y)^h)\,\mathrm{d}x\,\mathrm{d}y \\ &\quad\geq 0,\end{aligned}$$

where the last inequality follows from $d_U(y) - d_U(x) = d_W(x) - d_W(y)$ and the monotonicity of $t \mapsto t^h$ on $[0,1]$. Since $d_U = 1 - d_W$,

$$\begin{aligned}\int d_W(x)^h d_U(x)^2\,\mathrm{d}x &= \int d_W(x)^h (1 - d_W(x))^2\,\mathrm{d}x \\ &= M_h(W) - 2M_{h+1}(W) + M_{h+2}(W).\end{aligned}$$

Combining these estimates and using $q = 1 - p$, we obtain

$$\begin{aligned}
&\int d_W(x)^h \tau_W(x)\,\mathrm{d}x \\
&\geq 2\langle d_W^h, T_W d_W\rangle - pM_h(W) \\
&= 2M_{h+1}(W) - (2q+p)M_h(W) + 2\langle d_W^h, T_U d_U\rangle \\
&\geq 2M_{h+1}(W) - (2q+p)M_h(W) + 2\int d_W(x)^h d_U(x)^2\,\mathrm{d}x \\
&= 2M_{h+1}(W) - (2q+p)M_h(W) + 2\big(M_h(W) - 2M_{h+1}(W) + M_{h+2}(W)\big) \\
&= 2M_{h+2}(W) - 2M_{h+1}(W) + pM_h(W) \\
&= \left(2 - \frac{1}{p}\right) M_{h+2}(W) + \frac{1}{p}\big(M_{h+2}(W) - 2pM_{h+1}(W) + p^2 M_h(W)\big) \\
&= \left(2 - \frac{1}{p}\right) M_{h+2}(W) + \frac{1}{p}\int d_W(x)^h (d_W(x) - p)^2\,\mathrm{d}x \\
&\geq \left(2 - \frac{1}{p}\right) M_{h+2}(W).
\end{aligned}$$

This proves the claim. □

3.2 Adding a universal vertex

Proposition 3.2. *Let F have $n \geq 2$ vertices, and let $0 \leq \underline{p} < 1$. Suppose Φ_F is nondecreasing and convex on $[\underline{p}, 1]$, with $\Phi_F(\underline{p}) = 0$, and $t(F, V) \geq \Phi_F(z)$ for every graphon V of edge density $z > \underline{p}$. Then every graphon W of edge density $p > 1/(2 - \underline{p})$ satisfies*

$$t(K_1 \vee F, W) \geq \Phi_{K_1 \vee F}(p).$$

Moreover, $\Phi_{K_1\vee F}$ is nondecreasing and convex on $[1/(2-\underline{p}), 1]$, with $\Phi_{K_1\vee F}(1/(2-\underline{p})) = 0$. Thus $K_1 \vee F$ satisfies the hypotheses of the proposition with $1/(2-\underline{p})$ in place of $\underline{p}$.

Proof. For any $c \in [\underline{p}, 1]$, the tangent line to Φ_F at c gives

$$t(F, V) \geq \Phi_F(c) + \Phi_F'(c)(z - c), \qquad 0 \leq z \leq 1. \tag{3.2}$$

For $z > \underline{p}$, this follows from the assumed bound and convexity. For $z \leq \underline{p}$, nonnegativity of $\Phi_F'(c)$ and convexity give

$$\begin{aligned}
\Phi_F(c) + \Phi_F'(c)(z - c) &\leq \Phi_F(c) + \Phi_F'(c)(\underline{p} - c) \\
&\leq \Phi_F(\underline{p}) = 0 \leq t(F, V).
\end{aligned}$$

By definition,

$$t(K_1 \vee F, W) = \int \left(\int \prod_{uv \in E(F)} W(y_u, y_v) \prod_{v \in V(F)} W(x, y_v) \prod_{v \in V(F)} \mathrm{d}y_v \right) \mathrm{d}x.$$

Fix $x \in [0,1]$ with $d_W(x) > 0$, and define the map $Q_x : [0,1] \to [0,1]$ by

$$Q_x(s) := \inf\left\{ y \in [0,1] : \int_0^y W(x,z)\,\mathrm{d}z \geq s\, d_W(x) \right\}, \qquad s \in [0,1].$$

The map Q_x is nondecreasing and satisfies

$$\int \mathbf{1}_{\{Q_x(s) \leq y\}}\,\mathrm{d}s = \frac{1}{d_W(x)} \int_0^y W(x,z)\,\mathrm{d}z, \qquad y \in [0,1].$$

Consequently, for every bounded measurable $f : [0,1] \to \mathbb{R}$,

$$\int f(Q_x(s))\,\mathrm{d}s = \frac{1}{d_W(x)} \int f(y) W(x,y)\,\mathrm{d}y.$$

Define the reweighted graphon

$$W_x(s,t) = W(Q_x(s), Q_x(t)).$$

Applying the integration identity for Q_x successively to the variables $(s_v)_{v \in V(F)}$, we obtain

$$\begin{aligned}
t(F, W_x) &= \int \prod_{uv \in E(F)} W_x(s_u, s_v) \prod_{v \in V(F)} \mathrm{d}s_v \\
&= \int \prod_{uv \in E(F)} W(Q_x(s_u), Q_x(s_v)) \prod_{v \in V(F)} \mathrm{d}s_v \\
&= \int \prod_{uv \in E(F)} W(y_u, y_v) \left(\prod_{v \in V(F)} \frac{W(x, y_v)}{d_W(x)} \right) \prod_{v \in V(F)} \mathrm{d}y_v \\
&= \frac{1}{d_W(x)^n} \int \prod_{uv \in E(F)} W(y_u, y_v) \prod_{v \in V(F)} W(x, y_v) \prod_{v \in V(F)} \mathrm{d}y_v.
\end{aligned}$$

In particular, its edge density is

$$t(K_2, W_x) = \frac{1}{d_W(x)^2} \int W(y,z) W(x,y) W(x,z)\,\mathrm{d}y\,\mathrm{d}z = \frac{\tau_W(x)}{d_W(x)^2}.$$

Multiplying the expression for $t(F, W_x)$ by $d_W(x)^n$ and integrating in x gives

$$t(K_1 \vee F, W) = \int d_W(x)^n t(F, W_x)\,\mathrm{d}x.$$

When $d_W(x) = 0$, the contribution of x is zero; we set $W_x = 0$ there.

Lemma 3.1, with $h = n - 2$, gives

$$\begin{aligned}
\int d_W(x)^n t(K_2, W_x)\,\mathrm{d}x &= \int d_W(x)^{n-2} \tau_W(x)\,\mathrm{d}x \\
&\geq \left(2 - \frac{1}{p}\right) M_n(W).
\end{aligned}$$

For $p > 1/(2 - \underline{p})$, the point $2 - 1/p$ lies in $(\underline{p}, 1]$. Applying (3.2) at this point to each W_x and integrating gives

$$\begin{aligned}
&t(K_1 \vee F, W) \\
&\quad = \int d_W(x)^n t(F, W_x)\,\mathrm{d}x \\
&\quad \geq \int d_W(x)^n \left[\Phi_F\left(2 - \frac{1}{p}\right) + \Phi'_F\left(2 - \frac{1}{p}\right) \left(t(K_2, W_x) - \left(2 - \frac{1}{p}\right) \right) \right] \mathrm{d}x \\
&\quad = M_n(W) \Phi_F\left(2 - \frac{1}{p}\right) + \Phi'_F\left(2 - \frac{1}{p}\right) \left(\int d_W(x)^n t(K_2, W_x)\,\mathrm{d}x - \left(2 - \frac{1}{p}\right) M_n(W) \right) \\
&\quad \geq M_n(W) \Phi_F\left(2 - \frac{1}{p}\right) \\
&\quad \geq \left(\int d_W(x)\,\mathrm{d}x \right)^n \Phi_F\left(2 - \frac{1}{p}\right) \\
&\quad = p^n \Phi_F\left(2 - \frac{1}{p}\right).
\end{aligned}$$

Here we use $\Phi_F'(2-1/p), \Phi_F(2-1/p) \geq 0$ and the preceding bound on $\int d_W(x)^n t(K_2, W_x)\,\mathrm{d}x$; the last inequality follows from Jensen's inequality. Since $P_{K_1 \vee F}(x) = xP_F(x-1)$,

$$\Phi_{K_1 \vee F}(p) = p^n \Phi_F\left(2 - \frac{1}{p}\right).$$

Thus $K_1 \vee F$ also satisfies the lower bound for $p > 1/(2-\underline{p})$.

For the second assertion, differentiating gives

$$\begin{aligned}
\Phi'_{K_1 \vee F}(p) &= np^{n-1}\Phi_F\left(2 - \frac{1}{p}\right) + p^{n-2}\Phi'_F\left(2 - \frac{1}{p}\right), \\
\Phi''_{K_1 \vee F}(p) &= n(n-1)p^{n-2}\Phi_F\left(2 - \frac{1}{p}\right) + 2(n-1)p^{n-3}\Phi'_F\left(2 - \frac{1}{p}\right) \\
&\quad + p^{n-4}\Phi''_F\left(2 - \frac{1}{p}\right).
\end{aligned}$$

These are nonnegative on $[1/(2-\underline{p}), 1]$, since $2 - 1/p$ lies in $[\underline{p}, 1]$ there and $\Phi_F, \Phi'_F, \Phi''_F \geq 0$ on $[\underline{p}, 1]$. Finally,

$$\Phi_{K_1 \vee F}\left(\frac{1}{2-\underline{p}}\right) = \frac{\Phi_F(\underline{p})}{(2-\underline{p})^n} = 0.$$

□

Iterating Proposition 3.2 shows that, under its hypotheses, $t(K_s \vee F, W) \geq \Phi_{K_s \vee F}(p)$ for every integer $s \geq 0$ and every graphon W of edge density

$$p > 1 - \frac{1}{s + 1/(1-\underline{p})}. \tag{3.3}$$

Indeed, with $\underline{p}_0 = \underline{p}$, the thresholds obey $\underline{p}_{j+1} = 1/(2-\underline{p}_j)$, or equivalently $(1-\underline{p}_{j+1})^{-1} = (1-\underline{p}_j)^{-1} + 1$; by the second part of Proposition 3.2, the hypotheses are preserved at every step.

Proof of Corollary 1.5. For odd $\ell \geq 3$,

$$\Phi_{C_\ell}(p) = p\big(p^{\ell-1} - (1-p)^{\ell-1}\big).$$

This polynomial is nonnegative on $[1/2, 1]$, with $\Phi_{C_\ell}(1/2) = 0$. On the same interval, its derivatives satisfy

$$\begin{aligned}
\Phi'_{C_\ell}(p) &= p^{\ell-1} - (1-p)^{\ell-1} + (\ell-1)p\big(p^{\ell-2} + (1-p)^{\ell-2}\big) \geq 0, \\
\Phi''_{C_\ell}(p) &= 2(\ell-1)\big(p^{\ell-2} + (1-p)^{\ell-2}\big) \\
&\quad + (\ell-1)(\ell-2)p\big(p^{\ell-3} - (1-p)^{\ell-3}\big) \geq 0.
\end{aligned}$$

Apply Theorem 1.4 and Proposition 3.2 and (3.3) with $\underline{p} = 1/2$. Since $\chi(K_s \vee C_\ell) = s+3$, the resulting threshold is the required one. □

3.3 Trivially perfect graphs

Proof of Theorem 1.3. We use strong induction on $v(H)$, assuming that every trivially perfect graph F with at least one edge and $v(F) < v(H)$ satisfies the Goodman-type lower bound, with Φ_F nondecreasing and convex on $[\underline{p}_F, 1]$ and $\Phi_F(\underline{p}_F) = 0$.

The base case is a star H with $m \geq 1$ leaves. Here $\underline{p}_H = 0$, and Jensen's inequality gives

$$t(H, W) = \int d_W(x)^m\,\mathrm{d}x \geq p^m = \Phi_H(p).$$

The polynomial p^m is nondecreasing and convex on $[0,1]$ and equals zero at $p=0$.

If H is disconnected, write $H = H_1 \sqcup H_2$, where H_1, H_2 are nonempty. Since $\chi(H) = \max\{\chi(H_1), \chi(H_2)\}$,

$$t(H,W) = t(H_1,W)t(H_2,W) \geq \Phi_{H_1}(p)\Phi_{H_2}(p) = \Phi_H(p), \qquad p > \underline{p}_H.$$

Here $P_H = P_{H_1}P_{H_2}$ and $v(H) = v(H_1) + v(H_2)$ give the last equality; if H_i is edgeless, then $t(H_i,W) = \Phi_{H_i}(p) = 1$. The product rule gives nonnegativity of Φ'_H and Φ''_H on $[\underline{p}_H, 1]$. Relabelling so that $\chi(H_1) = \chi(H)$, we have

$$\Phi_H(\underline{p}_H) = \Phi_{H_1}(\underline{p}_{H_1})\Phi_{H_2}(\underline{p}_H) = 0.$$

It remains to consider $H = K_1 \vee F$, where F is trivially perfect and has an edge [2]. The induction hypothesis allows us to apply Proposition 3.2 with $\underline{p} = \underline{p}_F$. This gives the required bound and the required properties of Φ_H, since $\chi(H) = \chi(F) + 1$ and

$$\frac{1}{2 - \underline{p}_F} = 1 - \frac{1}{\chi(F)} = \underline{p}_H.$$

□

4 Eventual validity and girth

We first expand $t(H, 1-U) - \Phi_H(1-q)$ over forests, where $q = t(K_2, U)$. When the girth is odd, estimates for tree and cycle densities give the lower bound for small maximum degree. We then pass from small maximum degree to small edge density. The section ends with the obstruction for even girth and the proof of Theorem 1.6.

For a graphon U of edge density q, we use

$$\begin{aligned} d_U(x) &= \int U(x,y)\,\mathrm{d}y, & \Delta_U &= \|d_U\|_\infty, \\ M_j(U) &= \int d_U(x)^j\,\mathrm{d}x \quad (j \geq 0), & \mathrm{Var}(d_U) &= \int (d_U(x) - q)^2\,\mathrm{d}x = M_2(U) - q^2. \end{aligned} \tag{4.1}$$

Here $\mathrm{Var}(d_U)$ is the variance of the degree function, and the norm is the essential supremum. Thus $0 \leq q \leq \Delta_U \leq 1$.

4.1 Expansion over forests

For $A \subseteq E(H)$, let $H_A = (V(H), A)$ and define

$$r(A) = v(H) - \mathrm{comp}(H_A),$$

where $\mathrm{comp}(H_A)$ counts all connected components, including isolated vertices. Thus $r(A)$ is the number of edges in a spanning forest of H_A.

Lemma 4.1. *For every graph H and every graphon U of edge density q,*

$$\Phi_H(1-q) = \sum_{A \subseteq E(H)} (-1)^{|A|} q^{r(A)}, \tag{4.2}$$

$$t(H, 1-U) = \sum_{A \subseteq E(H)} (-1)^{|A|} t(H_A, U). \tag{4.3}$$

Proof. Whitney's spanning-subgraph expansion [19] follows by inclusion–exclusion over monochromatic edges:

$$P_H(k) = \sum_{A \subseteq E(H)} (-1)^{|A|} k^{\mathrm{comp}(H_A)} \qquad (k \in \mathbb{N}).$$

Both sides are polynomials in k, so the identity holds for all $k \in \mathbb{R}$. For $q > 0$, substituting $k = 1/q$ gives

$$\Phi_H(1-q) = q^{v(H)} P_H(1/q) = \sum_{A \subseteq E(H)} (-1)^{|A|} q^{v(H) - \mathrm{comp}(H_A)}.$$

Continuity gives the same formula at $q = 0$. Also,

$$\begin{aligned} t(H, 1-U) &= \int \prod_{uv \in E(H)} (1 - U(x_u, x_v)) \prod_{v \in V(H)} \mathrm{d}x_v \\ &= \sum_{A \subseteq E(H)} (-1)^{|A|} \int \prod_{uv \in A} U(x_u, x_v) \prod_{v \in V(H)} \mathrm{d}x_v \\ &= \sum_{A \subseteq E(H)} (-1)^{|A|} t(H_A, U). \end{aligned}$$

□

Subtracting the two identities gives

$$t(H, 1-U) - \Phi_H(1-q) = \sum_{A \subseteq E(H)} (-1)^{|A|} \big(t(H_A, U) - q^{r(A)} \big).$$

The summands with $r(A) = 0$ or $r(A) = 1$ are zero: the empty edge set contributes $1 - 1 = 0$, and each single edge contributes $-(q - q) = 0$.

Fix a total order $<$ on $E(H)$, and regard each forest $F \subseteq E(H)$ as the spanning graph $(V(H), F)$. Let $B(F)$ be the set of edges $e \in E(H) \setminus F$ such that adding e to F creates a cycle and every other edge of this cycle precedes e in the fixed order. Set

$$I_F(U) = \int \prod_{uv \in F} U(x_u, x_v) \prod_{uv \in B(F)} (1 - U(x_u, x_v)) \prod_{v \in V(H)} \mathrm{d}x_v.$$

Lemma 4.2 (Forest expansion)**.** *For every graphon U of edge density q,*

$$\Phi_H(1-q) = \sum_{\substack{F \text{ forest} \\ B(F) = \varnothing}} (-1)^{|F|} q^{|F|}, \tag{4.4}$$

$$t(H, 1-U) = \sum_{F \text{ forest}} (-1)^{|F|} I_F(U). \tag{4.5}$$

Here the sums range over forests $F \subseteq E(H)$.

Identity (4.4) is also known as Whitney's broken-circuit theorem [20], where a *broken circuit* is a cycle with its largest edge removed; $B(F) = \varnothing$ if and only if F contains no broken circuit.

Proof. We first show that every $A \subseteq E(H)$ has a unique representation

$$A = F \,\dot\cup\, D, \qquad F \text{ a forest}, \quad D \subseteq B(F).$$

Assign the weights $2, 4, \ldots, 2^{e(H)}$ to the edges of H in the fixed order. Given $A \subseteq E(H)$, let F be the minimum spanning forest of H_A. It is unique because different edge sets have different total weights. Adding an edge $e \in A \setminus F$ to F creates a cycle. Every other edge in this cycle has smaller weight than e: otherwise, replacing a heavier edge by e would give a spanning forest of smaller total weight. Hence $e \in B(F)$, giving the required representation with $D = A \setminus F$.

For uniqueness, suppose $A = F \dot\cup D$ with F a forest and $D \subseteq B(F)$. Since every edge of D has its endpoints in the same component of F, the graphs F and H_A have the same components. Let F' be the minimum spanning forest of H_A. If $F \neq F'$, choose $e \in F' \setminus F \subseteq D$. Adding e to F creates a cycle whose other edges all have smaller weight than e. As the weights are distinct powers of 2, the sum of their weights is smaller than the weight of e. Replacing e in F' by these other edges gives a graph with the same components as H_A and smaller total weight than F'. A spanning forest of this graph would also have smaller total weight than F', contradicting its minimality. Thus $F = F'$, so A determines F and then $D = A \setminus F$. We also have $r(A) = |F|$.

Using this decomposition in the expansion of $t(H, 1-U)$, we obtain

$$\begin{aligned} &t(H, 1-U) \\ &\quad= \sum_{F \text{ forest}} \sum_{D \subseteq B(F)} (-1)^{|F|+|D|} \int \prod_{uv \in F \dot\cup D} U(x_u, x_v) \prod_{v \in V(H)} \mathrm{d}x_v \\ &\quad= \sum_{F \text{ forest}} (-1)^{|F|} \int \prod_{uv \in F} U(x_u, x_v) \prod_{uv \in B(F)} (1 - U(x_u, x_v)) \prod_{v \in V(H)} \mathrm{d}x_v \\ &\quad= \sum_{F \text{ forest}} (-1)^{|F|} I_F(U). \end{aligned}$$

For each forest F, the corresponding terms in (4.2) sum to

$$(-1)^{|F|} q^{|F|} \sum_{D \subseteq B(F)} (-1)^{|D|} = \begin{cases} (-1)^{|F|} q^{|F|}, & B(F) = \varnothing, \\ 0, & B(F) \neq \varnothing. \end{cases}$$

Summing over F proves (4.4). □

Since $I_F(U) = t(F, U)$ when $B(F) = \varnothing$, subtracting the two identities gives

$$\begin{aligned} t(H, 1-U) - \Phi_H(1-q) = &\sum_{\substack{F \text{ forest} \\ B(F) = \varnothing}} (-1)^{|F|} (t(F, U) - q^{|F|}) \\ &+ \sum_{\substack{F \text{ forest} \\ B(F) \neq \varnothing}} (-1)^{|F|} I_F(U). \end{aligned} \tag{4.6}$$

4.2 Tree and forest densities

The lower bound below is the tree case of Sidorenko's inequality [15]; the upper bound is his comparison with the star having the same number of edges [16].

Lemma 4.3 (Tree moment bounds)**.** *If T is a tree with $r \geq 1$ edges, then*

$$q^r \leq t(T, U) \leq M_r(U). \tag{4.7}$$

Sidorenko's inequality for trees holds with equality whenever U is regular, that is, $d_U(x) = q$ almost everywhere. The following lemma bounds the gap $t(F, U) - q^{e(F)}$ for forests F in terms of the variance of the degree function.

Lemma 4.4 (Variance estimates)**.** *If F is a forest with $r \geq 2$ edges, then*

$$0 \leq t(F,U) - q^r \leq \binom{r}{2} \Delta_U^{r-2} \operatorname{Var}(d_U). \tag{4.8}$$

In particular, taking F to be the star $K_{1,r}$ gives

$$0 \leq M_r(U) - q^r \leq \binom{r}{2} \Delta_U^{r-2} \operatorname{Var}(d_U). \tag{4.9}$$

Proof. Assume $\Delta_U > 0$, since the case $\Delta_U = 0$ is immediate. Let $T_1, \dots, T_\ell$ be the components of F containing an edge, and write $r_i = e(T_i)$, so that $\sum_i r_i = r$. By Lemma 4.3 and Hölder's inequality,

$$\begin{aligned} q^r = \prod_{i=1}^{\ell} q^{r_i} \leq t(F,U) &= \prod_{i=1}^{\ell} t(T_i,U) \\ &\leq \prod_{i=1}^{\ell} M_{r_i}(U) \leq \prod_{i=1}^{\ell} M_r(U)^{r_i/r} = M_r(U). \end{aligned}$$

On $[0, \Delta_U]$, the function $s \mapsto s^r$ has second derivative at most $r(r-1)\Delta_U^{r-2}$. Since $q, d_U(x) \in [0, \Delta_U]$ for almost every x, Taylor's theorem at $s = q$ gives

$$d_U(x)^r \leq q^r + r q^{r-1}(d_U(x) - q) + \binom{r}{2} \Delta_U^{r-2} (d_U(x) - q)^2.$$

Integrating and using $\int (d_U(x) - q)\,\mathrm{d}x = 0$, we obtain

$$0 \leq t(F,U) - q^r \leq M_r(U) - q^r \leq \binom{r}{2} \Delta_U^{r-2} \operatorname{Var}(d_U).$$

The star case follows from $t(K_{1,r}, U) = M_r(U)$. □

4.3 Path and cycle densities

We bound the difference between the densities of a path and the cycle obtained by joining its endpoints.

Lemma 4.5 (A path estimate)**.** *Let F be a forest with r edges, and let $u, v \in V(F)$ have distance $s - 1 \geq 2$. Then*

$$\begin{aligned} 0 &\leq \int \Big(\prod_{ab \in E(F)} U(x_a, x_b) \Big) (1 - U(x_u, x_v)) \prod_{w \in V(F)} \mathrm{d}x_w \\ &\leq \Delta_U^{r-s+1} \big(t(P_s, U) - t(C_s, U) \big). \end{aligned} \tag{4.10}$$

Proof. Write the u–v path as $u = v_1, \dots, v_s = v$. If a is a leaf outside this path with neighbour b, then x_a occurs only in the factor $U(x_a, x_b)$. Hence

$$\begin{aligned} &\int \Big(\prod_{cd \in E(F)} U(x_c, x_d) \Big) (1 - U(x_u, x_v))\, \mathrm{d}x_a \\ &\quad = d_U(x_b) \Big(\prod_{cd \in E(F) \setminus \{ab\}} U(x_c, x_d) \Big) (1 - U(x_u, x_v)) \\ &\quad \leq \Delta_U \Big(\prod_{cd \in E(F) \setminus \{ab\}} U(x_c, x_d) \Big) (1 - U(x_u, x_v)). \end{aligned}$$

Repeating this for the $r-s+1$ edges outside the path and integrating isolated vertices, which contribute 1, gives

$$\begin{aligned}
0 &\le \int \Big(\prod_{ab\in E(F)} U(x_a,x_b) \Big)(1-U(x_u,x_v)) \prod_{w\in V(F)} \mathrm{d}x_w \\
&\le \Delta_U^{r-s+1} \int \Big(\prod_{i=1}^{s-1} U(x_{v_i},x_{v_{i+1}}) \Big)(1-U(x_u,x_v)) \prod_{i=1}^{s} \mathrm{d}x_{v_i} \\
&= \Delta_U^{r-s+1} \big(t(P_s,U)-t(C_s,U)\big).
\end{aligned}$$

□

For every $s \ge 3$,

$$t(P_s,U)-t(C_s,U) = \int \Big(\prod_{i=1}^{s-1} U(x_i,x_{i+1}) \Big)(1-U(x_1,x_s)) \prod_{i=1}^{s} \mathrm{d}x_i.$$

For $s=3$,

$$\begin{aligned}
M_2(U)-t(C_3,U) &= t(P_3,U)-t(C_3,U) \\
&= \int U(x,y)U(y,z)(1-U(x,z))\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z.
\end{aligned} \tag{4.11}$$

Lemma 4.6. *For every integer $s \ge 3$,*

$$0 \le t(P_s,U)-t(C_s,U) \le (s-2)\Delta_U^{s-3}\big(M_2(U)-t(C_3,U)\big). \tag{4.12}$$

Proof. For $s=3$, the claim is immediate by (4.11).

Define the iterated kernels recursively by

$$U^{\circ 1}(x,y) = U(x,y), \qquad U^{\circ(\ell+1)}(x,y) = \int U^{\circ\ell}(x,z)U(z,y)\,\mathrm{d}z.$$

Induction gives

$$0 \le U^{\circ\ell}(x,y) \le \Delta_U^{\ell-1} \quad \text{for almost every } (x,y), \tag{4.13}$$

since $U \le 1$ and

$$U^{\circ(\ell+1)}(x,y) \le \Delta_U^{\ell-1} \int U(z,y)\,\mathrm{d}z = \Delta_U^{\ell-1} d_U(y) \le \Delta_U^{\ell}.$$

We obtain

$$\begin{aligned}
&t(P_s,U)-t(C_s,U) \\
&\quad= \int \Big(\prod_{i=1}^{s-1} U(x_i,x_{i+1}) \Big)(1-U(x_1,x_s)) \prod_{i=1}^{s} \mathrm{d}x_i \\
&\quad\le \int \Big(\prod_{i=1}^{s-1} U(x_i,x_{i+1}) \Big)(1-U(x_1,x_{s-1})) \prod_{i=1}^{s} \mathrm{d}x_i \\
&\qquad + \int \Big(\prod_{i=1}^{s-1} U(x_i,x_{i+1}) \Big) U(x_1,x_{s-1})(1-U(x_1,x_s)) \prod_{i=1}^{s} \mathrm{d}x_i \\
&\quad= \int \Big(\prod_{i=1}^{s-2} U(x_i,x_{i+1}) \Big) d_U(x_{s-1})(1-U(x_1,x_{s-1})) \prod_{i=1}^{s-1} \mathrm{d}x_i \\
&\qquad + \int U^{\circ(s-2)}(x,y)U(x,y)U(y,z)(1-U(x,z))\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \\
&\quad\le \Delta_U\big(t(P_{s-1},U)-t(C_{s-1},U)\big) + \Delta_U^{s-3}\big(M_2(U)-t(C_3,U)\big),
\end{aligned}$$

where the first inequality follows from

$$1 - U(x_1, x_s) \le 1 - U(x_1, x_{s-1}) + U(x_1, x_{s-1})(1 - U(x_1, x_s)),$$

and the last follows from $d_U \le \Delta_U$, (4.13), and (4.11). In the second equality, we integrate x_s in the first term and $x_2, \dots, x_{s-2}$ in the second. Induction on s proves (4.12). □

4.4 Comparison of cycle densities

For odd $g \ge 5$, the following comparison provides the estimate needed for cycles of length at least g.

Theorem 4.7. *Let $g \ge 5$ be odd and let $s \ge g$ be an integer. For every graphon U,*

$$\begin{aligned} 0 &\le M_{s-1}(U) - t(C_s, U) \\ &\le \frac{(s-2)(s-3)}{(g-2)(g-3)} \, \Delta_U^{s-g} (M_{g-1}(U) - t(C_g, U)). \end{aligned} \tag{4.14}$$

Lemma 4.8 (Scalar moment comparison)**.** *Let $2 \le a \le b$ be integers and let $\Delta > 0$. If a random variable Z takes values in $[0, \Delta]$ and $0 \le \alpha \le \mathbb{E}Z$, then*

$$\mathbb{E}Z^b - \alpha^b \le \frac{b(b-1)}{a(a-1)} \Delta^{b-a} (\mathbb{E}Z^a - \alpha^a). \tag{4.15}$$

Proof. Equality holds when $a = b$. For $b > a$, consider the polynomial

$$z^b - \frac{b(b-1)}{a(a-1)} \Delta^{b-a} z^a.$$

Its first and second derivatives on $[0, \Delta]$ are, respectively,

$$\begin{aligned} b z^{a-1} \left(z^{b-a} - \frac{b-1}{a-1} \Delta^{b-a} \right) &\le 0, \\ b(b-1) z^{a-2} (z^{b-a} - \Delta^{b-a}) &\le 0. \end{aligned}$$

The polynomial is therefore nonincreasing and concave. Jensen's inequality and $\alpha \le \mathbb{E}Z$ give

$$\begin{aligned} \mathbb{E}Z^b - \frac{b(b-1)}{a(a-1)} \Delta^{b-a} \mathbb{E}Z^a &\le (\mathbb{E}Z)^b - \frac{b(b-1)}{a(a-1)} \Delta^{b-a} (\mathbb{E}Z)^a \\ &\le \alpha^b - \frac{b(b-1)}{a(a-1)} \Delta^{b-a} \alpha^a. \end{aligned}$$

Rearranging proves the claim. □

We next state the spectral identities for the integral operator T_U; see Lovász [9, Chapter 7].

Lemma 4.9 (Spectral identities)**.** *Let $(T_U f)(x) = \int U(x,y) f(y) \, \mathrm{d}y$ on real $L^2[0,1]$. Choose real orthonormal eigenfunctions f_i corresponding to all nonzero eigenvalues λ_i of T_U, counted with multiplicity. Then*

$$|\lambda_i| \le \int d_U(x) f_i(x)^2 \, \mathrm{d}x \le \Delta_U, \qquad \sum_i \lambda_i^2 = \int U(x,y)^2 \, \mathrm{d}x \, \mathrm{d}y. \tag{4.16}$$

For every integer $s \ge 3$ and every integer $j \ge 1$,

$$t(C_s, U) = \sum_i \lambda_i^s, \tag{4.17}$$

$$\sum_i \lambda_i^2 \int d_U(x)^j f_i(x)^2 \, \mathrm{d}x = \int d_U(x)^j U(x,y)^2 \, \mathrm{d}x \, \mathrm{d}y. \tag{4.18}$$

All these series converge absolutely.

Proof. The operator T_U is compact, self-adjoint, and Hilbert–Schmidt. The spectral theorem gives the eigenfunctions f_i; adjoining an orthonormal basis of $\ker T_U$ gives a complete eigenbasis. For every real $f \in L^2[0,1]$, symmetry gives

$$\int d_U(x) f(x)^2 \,\mathrm{d}x \pm \langle f, T_U f\rangle = \frac{1}{2}\int U(x,y)(f(x) \pm f(y))^2 \,\mathrm{d}x\,\mathrm{d}y \geq 0. \tag{4.19}$$

Taking $f = f_i$ proves the eigenvalue bound. The formula for $\sum_i \lambda_i^2$ is the Hilbert–Schmidt norm identity.

For $s \geq 3$, the operators T_U and T_U^{s-1} are both Hilbert–Schmidt, and $U^{\circ(s-1)}$ is the kernel of T_U^{s-1}. Computing their inner product from the kernels and the eigenbasis gives

$$t(C_s, U) = \int U(x,y) U^{\circ(s-1)}(x,y) \,\mathrm{d}x\,\mathrm{d}y = \langle T_U, T_U^{s-1}\rangle_{\mathrm{HS}} = \sum_i \lambda_i^s.$$

For almost every x, Parseval's identity also gives

$$\int U(x,y)^2 \,\mathrm{d}y = \sum_i \left(\int U(x,y) f_i(y) \,\mathrm{d}y\right)^2 = \sum_i \lambda_i^2 f_i(x)^2.$$

Multiplying by $d_U(x)^j$ and integrating proves (4.18) by Tonelli's theorem. Absolute convergence follows from $|\lambda_i| \leq \Delta_U \leq 1$ and $\sum_i \lambda_i^2 < \infty$. □

Lemma 4.10 (Nonnegative decomposition)**.** *With the eigenfunctions and eigenvalues above, for every integer $s \geq 3$,*

$$\begin{aligned} M_{s-1}(U) - t(C_s, U) = {} & \int d_U(x)^{s-2} U(x,y)(1 - U(x,y)) \,\mathrm{d}x\,\mathrm{d}y \\ & + \sum_i \lambda_i^2 \left(\int d_U(x)^{s-2} f_i(x)^2 \,\mathrm{d}x - \lambda_i^{s-2}\right). \end{aligned} \tag{4.20}$$

The integral and each summand are nonnegative, and the series converges absolutely.

Proof. By (4.18),

$$\begin{aligned} M_{s-1}(U) &= \int d_U(x)^{s-2} U(x,y) \,\mathrm{d}x\,\mathrm{d}y \\ &= \int d_U(x)^{s-2} U(x,y)(1 - U(x,y)) \,\mathrm{d}x\,\mathrm{d}y \\ &\quad + \sum_i \lambda_i^2 \int d_U(x)^{s-2} f_i(x)^2 \,\mathrm{d}x. \end{aligned}$$

Subtracting (4.17) gives the decomposition. Since $\int f_i(x)^2 \,\mathrm{d}x = 1$, Jensen's inequality and (4.16) give, for $j \geq 1$,

$$\int d_U(x)^j f_i(x)^2 \,\mathrm{d}x \geq \left(\int d_U(x) f_i(x)^2 \,\mathrm{d}x\right)^j \geq |\lambda_i|^j \geq \lambda_i^j.$$

Thus the integral and each summand are nonnegative. For absolute convergence, we have

$$\begin{aligned} & \sum_i \lambda_i^2 \left|\int d_U(x)^{s-2} f_i(x)^2 \,\mathrm{d}x - \lambda_i^{s-2}\right| \\ & \quad \leq \sum_i \lambda_i^2 \left(\int d_U(x)^{s-2} f_i(x)^2 \,\mathrm{d}x + |\lambda_i|^{s-2}\right) \\ & \quad \leq 2\Delta_U^{s-2} \sum_i \lambda_i^2 = 2\Delta_U^{s-2} \int U(x,y)^2 \,\mathrm{d}x\,\mathrm{d}y < \infty, \end{aligned}$$

where we used $\int f_i(x)^2 \,\mathrm{d}x = 1$, $d_U \leq \Delta_U$, and (4.16). □

Proof of Theorem 4.7. If $\Delta_U = 0$, then U is regular of degree zero, so $U = 0$ almost everywhere and both sides of (4.14) are zero. We may therefore assume $\Delta_U > 0$.

Fix i with $\lambda_i \geq 0$. Since $\int f_i(x)^2 \,\mathrm{d}x = 1$, let X have probability density $f_i(x)^2$ on $[0,1]$, and set $Z = d_U(X)$. Then $0 \leq Z \leq \Delta_U$ almost surely, and

$$\mathbb{E}Z^j = \int d_U(x)^j f_i(x)^2 \,\mathrm{d}x \qquad (j \geq 1).$$

Moreover, (4.16) gives

$$0 \leq \lambda_i \leq \int d_U(x) f_i(x)^2 \,\mathrm{d}x = \mathbb{E}Z.$$

Since $2 \leq g-2 \leq s-2$, we can apply Lemma 4.8 with

$$a = g-2, \qquad b = s-2, \qquad \alpha = \lambda_i, \qquad \Delta = \Delta_U.$$

This gives

$$\begin{aligned} &\int d_U(x)^{s-2} f_i(x)^2 \,\mathrm{d}x - \lambda_i^{s-2} \\ &\quad \leq \frac{(s-2)(s-3)}{(g-2)(g-3)} \Delta_U^{s-g} \left(\int d_U(x)^{g-2} f_i(x)^2 \,\mathrm{d}x - \lambda_i^{g-2} \right). \end{aligned} \tag{4.21}$$

For $\lambda_i < 0$, use that $g-2$ is odd to obtain

$$\begin{aligned} &\int d_U(x)^{s-2} f_i(x)^2 \,\mathrm{d}x - \lambda_i^{s-2} \\ &\quad \leq \int d_U(x)^{s-2} f_i(x)^2 \,\mathrm{d}x + |\lambda_i|^{s-2} \\ &\quad \leq \Delta_U^{s-g} \left(\int d_U(x)^{g-2} f_i(x)^2 \,\mathrm{d}x + |\lambda_i|^{g-2} \right) \\ &\quad = \Delta_U^{s-g} \left(\int d_U(x)^{g-2} f_i(x)^2 \,\mathrm{d}x - \lambda_i^{g-2} \right). \end{aligned} \tag{4.22}$$

By (4.20),

$$\begin{aligned} &M_{s-1}(U) - t(C_s, U) \\ &\quad = \int d_U(x)^{s-2} U(x,y)(1 - U(x,y)) \,\mathrm{d}x \,\mathrm{d}y \\ &\qquad + \sum_i \lambda_i^2 \left(\int d_U(x)^{s-2} f_i(x)^2 \,\mathrm{d}x - \lambda_i^{s-2} \right) \\ &\quad \leq \Delta_U^{s-g} \int d_U(x)^{g-2} U(x,y)(1 - U(x,y)) \,\mathrm{d}x \,\mathrm{d}y \\ &\qquad + \frac{(s-2)(s-3)}{(g-2)(g-3)} \Delta_U^{s-g} \sum_{i:\lambda_i \geq 0} \lambda_i^2 \left(\int d_U(x)^{g-2} f_i(x)^2 \,\mathrm{d}x - \lambda_i^{g-2} \right) \\ &\qquad + \Delta_U^{s-g} \sum_{i:\lambda_i < 0} \lambda_i^2 \left(\int d_U(x)^{g-2} f_i(x)^2 \,\mathrm{d}x - \lambda_i^{g-2} \right) \\ &\quad \leq \frac{(s-2)(s-3)}{(g-2)(g-3)} \Delta_U^{s-g} \Bigg(\int d_U(x)^{g-2} U(x,y)(1 - U(x,y)) \,\mathrm{d}x \,\mathrm{d}y \\ &\qquad\qquad + \sum_i \lambda_i^2 \left(\int d_U(x)^{g-2} f_i(x)^2 \,\mathrm{d}x - \lambda_i^{g-2} \right) \Bigg) \\ &\quad = \frac{(s-2)(s-3)}{(g-2)(g-3)} \Delta_U^{s-g} (M_{g-1}(U) - t(C_g, U)). \end{aligned}$$

The first inequality follows from (4.21), (4.22), and $d_U(x)^{s-2} \leq \Delta_U^{s-g} d_U(x)^{g-2}$. The second follows from $(s-2)(s-3)/((g-2)(g-3)) \geq 1$ and the nonnegativity of the integral and summands. The last equality is (4.20) with $s = g$. □

4.5 The bound for small maximum degree

For $s \geq 3$, let $c_s(H)$ be the number of simple s-cycles in H, counted without a choice of orientation or starting vertex.

In this subsection, H has girth g, where g is odd. Thus $c_g(H) \geq 1$ and $c_s(H) = 0$ for $s < g$.

Lemma 4.11 (Cycle count)**.** *For $g \leq s \leq v(H)$,*

$$2s\, c_s(H) \leq v(H)^{\lceil 2s/(g-1) \rceil} \leq v(H)^{s-g+3}. \tag{4.23}$$

In particular, $c_g(H) \leq v(H)^3$.

Proof. Set $h = (g-1)/2$ and $k = \lceil s/h \rceil$. Two distinct paths P, Q of length at most h with the same endpoints would contain a cycle in their union, giving

$$g \leq e(P \cup Q) \leq e(P) + e(Q) \leq 2h = g - 1.$$

Since $1 \leq s - (k-1)h \leq h$, the map

$$(v_0, \ldots, v_{s-1}) \longmapsto (v_0, v_h, \ldots, v_{(k-1)h})$$

is injective on oriented s-cycles with a specified starting vertex. Consequently,

$$2s\, c_s(H) \leq |V(H)^k| = v(H)^k.$$

Finally, $s \geq g = 2h + 1$ and $h \geq 1$ give

$$k = \left\lceil \frac{2s}{g-1} \right\rceil = 2 + \left\lceil \frac{s-g+1}{h} \right\rceil \leq 2 + (s - g + 1) = s - g + 3.$$

□

Deleting an edge of C_s and applying Lemma 4.3 gives

$$0 \leq t(P_s, U) - t(C_s, U) \leq M_{s-1}(U) - t(C_s, U) \qquad (s \geq 3). \tag{4.24}$$

For $g \leq s \leq v(H)$, we obtain

$$t(P_s, U) - t(C_s, U) \leq \begin{cases} M_{g-1}(U) - t(C_g, U), & s = g, \\ v(H)^2 \Delta_U^{s-g} \big(M_{g-1}(U) - t(C_g, U)\big), & g < s \leq v(H). \end{cases} \tag{4.25}$$

The case $s = g$ follows from (4.24). For $g = 3$ and $3 < s \leq v(H)$,

$$\begin{aligned} t(P_s, U) - t(C_s, U) &\leq (s-2)\Delta_U^{s-3}\big(M_2(U) - t(C_3, U)\big) \\ &\leq v(H)^2 \Delta_U^{s-3}\big(M_2(U) - t(C_3, U)\big), \end{aligned}$$

where the first inequality is Lemma 4.6 and the last follows from $s - 2 \leq v(H)^2$. For odd $g \geq 5$ and $g < s \leq v(H)$,

$$\begin{aligned} t(P_s, U) - t(C_s, U) &\leq M_{s-1}(U) - t(C_s, U) \\ &\leq \frac{(s-2)(s-3)}{(g-2)(g-3)} \Delta_U^{s-g}\big(M_{g-1}(U) - t(C_g, U)\big) \\ &\leq v(H)^2 \Delta_U^{s-g}\big(M_{g-1}(U) - t(C_g, U)\big), \end{aligned}$$

where the first two inequalities follow from (4.24) and Theorem 4.7, respectively, and the last follows from $(s-2)(s-3)/((g-2)(g-3)) \leq s^2 \leq v(H)^2$.

Proposition 4.12. *Let H have girth g, where g is odd. If $e(H)\Delta_U \le 1/2$ and $v(H)\Delta_U \le 1/2$, then*

$$\begin{aligned} &t(H,1-U)-\Phi_H(1-q)\\ &\quad\ge \left(\sum_{v\in V(H)}\binom{d_H(v)}{2}-c_3(H)-2v(H)^6\Delta_U\right)\mathrm{Var}(d_U)\\ &\qquad + c_g(H)(1-5v(H)^6\Delta_U)\big(M_{g-1}(U)-t(C_g,U)\big). \end{aligned} \tag{4.26}$$

In particular, $\Delta_U \le (6v(H)^6)^{-1}$ implies $t(H,1-U)\ge \Phi_H(1-q)$.

Proof. Each triangle contains three adjacent edge pairs, and an adjacent pair belongs to at most one triangle. Since H contains a cycle,

$$\sum_{v\in V(H)}\binom{d_H(v)}{2}-c_3(H)\ge 1.$$

Assume $\Delta_U > 0$.

For forests F with at most two edges,

$$t(F,U)=\begin{cases}1, & |F|=0,\\ q, & |F|=1,\\ q^2, & F \text{ has two disjoint edges},\\ \displaystyle\int d_U(x)^2\,\mathrm{d}x = M_2(U), & F \text{ has two adjacent edges}.\end{cases}$$

Their contribution to the first sum of (4.6) is

$$\begin{aligned} &\sum_{\substack{F \text{ forest},\, B(F)=\varnothing\\ |F|\le 2}}(-1)^{|F|}\big(t(F,U)-q^{|F|}\big)\\ &\quad=\left(\sum_{v\in V(H)}\binom{d_H(v)}{2}-c_3(H)\right)(M_2(U)-q^2)\\ &\quad=\left(\sum_{v\in V(H)}\binom{d_H(v)}{2}-c_3(H)\right)\mathrm{Var}(d_U). \end{aligned}$$

Here $\sum_{v\in V(H)}\binom{d_H(v)}{2}$ counts the pairs of adjacent edges, and exactly $c_3(H)$ of these have nonempty $B(F)$, one for each triangle with its largest edge removed.

If $B(F)\ne\varnothing$, choose $e=uv\in B(F)$. The edge e and the u–v path in F form a cycle, so $|F|\ge g-1$. If $|F|=g-1$, this cycle has length g, and F is obtained by deleting its largest edge e. Any other edge in $B(F)$ would join two vertices at distance at most $g-2$ along the path and create a cycle of length at most $g-1$. Thus $B(F)=\{e\}$, and

$$\begin{aligned} I_F(U)&=\int\Big(\prod_{ab\in F}U(x_a,x_b)\Big)(1-U(x_u,x_v))\prod_{w\in V(H)}\mathrm{d}x_w\\ &=t(P_g,U)-t(C_g,U). \end{aligned}$$

Each g-cycle gives exactly one such forest by deleting its largest edge. Hence

$$\begin{aligned} &\sum_{\substack{F \text{ forest},\, B(F)\ne\varnothing\\ |F|=g-1}}(-1)^{|F|}I_F(U)\\ &\quad=(-1)^{g-1}c_g(H)\big(t(P_g,U)-t(C_g,U)\big)\\ &\quad=c_g(H)\big(t(P_g,U)-t(C_g,U)\big), \end{aligned}$$

where the last equality follows from g being odd. Combining these contributions in (4.6) and bounding the remaining terms in absolute value gives

$$\begin{aligned} &t(H,1-U)-\Phi_H(1-q)\\ &\quad\ge\left(\sum_{v\in V(H)}\binom{d_H(v)}{2}-c_3(H)\right)\operatorname{Var}(d_U)+c_g(H)\big(t(P_g,U)-t(C_g,U)\big)\\ &\qquad-\sum_{\substack{F\text{ forest},\,B(F)=\varnothing\\ |F|\ge 3}}\big(t(F,U)-q^{|F|}\big)-\sum_{\substack{F\text{ forest},\,B(F)\neq\varnothing\\ |F|\ge g}}I_F(U). \end{aligned} \tag{4.27}$$

By (4.8),

$$\begin{aligned} &\sum_{\substack{F\text{ forest},\,B(F)=\varnothing\\ |F|\ge 3}}\big(t(F,U)-q^{|F|}\big)\\ &\quad\le\operatorname{Var}(d_U)\sum_{r=3}^{e(H)}\binom{e(H)}{r}\binom{r}{2}\Delta_U^{r-2}\\ &\quad=\binom{e(H)}{2}\big((1+\Delta_U)^{e(H)-2}-1\big)\operatorname{Var}(d_U)\le 2e(H)^3\Delta_U\operatorname{Var}(d_U). \end{aligned} \tag{4.28}$$

Here $e(H)\Delta_U\le 1/2$ and comparison with a geometric series give

$$(1+\Delta_U)^{e(H)}\le 2,\qquad (1+\Delta_U)^{e(H)}-1\le 2e(H)\Delta_U.$$

For each $e=uv\in B(F)$, Lemma 4.5 gives

$$I_F(U)\le\int\prod_{ab\in F}U(x_a,x_b)(1-U(x_u,x_v))\prod_{w\in V(H)}\mathrm{d}x_w\le\Delta_U^j\big(t(P_s,U)-t(C_s,U)\big),$$

where s is the length of the cycle formed by e and its path in F, and $j=|F|-s+1$. Here e is the largest edge of the cycle, and there are at most $\binom{e(H)}{j}$ choices for the remaining edges of F. For $|F|\ge g$, we have $j\ge 1$ when $s=g$ and $j\ge 0$ when $s>g$. Summing these bounds over all such forests F and all edges $e\in B(F)$ gives

$$\begin{aligned} &\sum_{\substack{F\text{ forest},\,B(F)\neq\varnothing\\ |F|\ge g}}I_F(U)\\ &\quad\le c_g(H)\big((1+\Delta_U)^{e(H)}-1\big)\big(t(P_g,U)-t(C_g,U)\big)\\ &\qquad+(1+\Delta_U)^{e(H)}\sum_{s=g+1}^{v(H)}c_s(H)\big(t(P_s,U)-t(C_s,U)\big). \end{aligned} \tag{4.29}$$

By (4.23) and (4.25),

$$\begin{aligned} &\sum_{s=g+1}^{v(H)}c_s(H)\big(t(P_s,U)-t(C_s,U)\big)\\ &\quad\le v(H)^5\big(M_{g-1}(U)-t(C_g,U)\big)\sum_{j=1}^{v(H)-g}(v(H)\Delta_U)^j\\ &\quad\le 2v(H)^6\Delta_U\big(M_{g-1}(U)-t(C_g,U)\big). \end{aligned}$$

It follows that

$$\begin{aligned}
&\sum_{\substack{F \text{ forest},\, B(F)\neq\varnothing \\ |F|\geq g}} I_F(U) \\
&\quad\leq (2c_g(H)e(H) + 4v(H)^6)\Delta_U\big(M_{g-1}(U) - t(C_g,U)\big) \\
&\quad\leq 5c_g(H)v(H)^6\Delta_U\big(M_{g-1}(U) - t(C_g,U)\big),
\end{aligned} \tag{4.30}$$

where $c_g(H) \geq 1$ and $2e(H) \leq v(H)^2 \leq v(H)^6$.

For $g = 3$, we have $t(P_g, U) = M_{g-1}(U)$. For $g \geq 5$, Lemma 4.3 and (4.9) give

$$\begin{aligned}
0 \leq M_{g-1}(U) - t(P_g,U) &\leq M_{g-1}(U) - q^{g-1} \\
&\leq \binom{g-1}{2}\Delta_U^{g-3}\operatorname{Var}(d_U).
\end{aligned}$$

Using $c_g(H) \leq v(H)^3$ and $\Delta_U \leq 1$, both cases give

$$c_g(H)\big(t(P_g,U) - t(C_g,U)\big) \geq c_g(H)\big(M_{g-1}(U) - t(C_g,U)\big) - \frac{v(H)^5}{2}\Delta_U^2\operatorname{Var}(d_U).$$

Combining this with (4.27), (4.28), and (4.30) gives (4.26), since

$$2e(H)^3\Delta_U + \frac{v(H)^5}{2}\Delta_U^2 \leq \left(\frac{v(H)^6}{4} + \frac{v(H)^5}{2}\right)\Delta_U \leq 2v(H)^6\Delta_U.$$

Finally, $\Delta_U \leq (6v(H)^6)^{-1}$ satisfies both hypotheses and gives

$$\begin{aligned}
&t(H,1-U) - \Phi_H(1-q) \\
&\quad\geq \frac{2}{3}\operatorname{Var}(d_U) + \frac{c_g(H)}{6}\big(M_{g-1}(U) - t(C_g,U)\big) \geq 0.
\end{aligned}$$

□

4.6 From maximum degree to edge density

For a graphon V, define

$$D_H(V) = t(H, 1-V) - 1 + e(H)\,t(K_2, V).$$

We write $(a)_+ = \max\{a, 0\}$.

Lemma 4.13. *For every graphon V, $D_H(V) \geq 0$. If V_1, V_2 are graphons with $V_1 + V_2 \leq 1$, then*

$$D_H(V_1 + V_2) \geq D_H(V_1) + D_H(V_2). \tag{4.31}$$

Proof. Choose independent uniform positions $(Z_v)_{v\in V(H)}$ in $[0,1]$. Conditional on these positions, select each edge uv independently with probability $V(Z_u, Z_v)$, and let N be the number of selected edges. Then $\mathbb{E}N = e(H)\,t(K_2, V)$ and $\mathbb{P}(N = 0) = t(H, 1-V)$, so

$$\begin{aligned}
D_H(V) &= \mathbb{E}N - 1 + \mathbb{P}(N=0) \\
&= \mathbb{E}\big(N - 1 + \mathbf{1}_{\{N=0\}}\big) = \mathbb{E}(N-1)_+ \geq 0.
\end{aligned} \tag{4.32}$$

To prove (4.31), assign each edge uv one of three outcomes with probabilities $V_1(Z_u, Z_v)$, $V_2(Z_u, Z_v)$, and $1 - V_1(Z_u, Z_v) - V_2(Z_u, Z_v)$, independently conditional on the positions. Let N_1 and N_2 be the numbers of edges with the first and second outcomes, respectively. Then

$$(N_1 + N_2 - 1)_+ \geq (N_1 - 1)_+ + (N_2 - 1)_+.$$

Taking expectations proves (4.31). □

Lemma 4.14. *For $e(H) \geq 1$ and $q_1, q_2 \geq 0$ with $q = q_1 + q_2 \leq 1/(2e(H))$,*

$$|\Phi_H(1-q) - \Phi_H(1-q_2) + e(H)q_1| \leq 2e(H)^2 q q_1. \tag{4.33}$$

Proof. If $e(H) = 1$, then $\Phi_H(p) = p$ and the left-hand side is zero. For $e(H) \geq 2$, (4.4) gives $\Phi'_H(1) = e(H)$ and shows that the coefficient of q^j in $\Phi_H(1-q)$ has absolute value at most $\binom{e(H)}{j}$. Hence, for $0 \leq x \leq 1/(2e(H))$,

$$\begin{aligned} |\Phi''_H(1-x)| &\leq \sum_{j=2}^{e(H)} \binom{e(H)}{j} j(j-1)x^{j-2} \\ &= e(H)(e(H)-1)(1+x)^{e(H)-2} \leq 2e(H)^2. \end{aligned}$$

Integrating gives

$$|e(H) - \Phi'_H(1-x)| = \left| \int_0^x \Phi''_H(1-y)\,\mathrm{d}y \right| \leq 2e(H)^2 x.$$

Since $q - q_2 = q_1$, a second integration gives

$$\begin{aligned} &|\Phi_H(1-q) - \Phi_H(1-q_2) + e(H)q_1| \\ &\quad = \left| \int_{q_2}^{q} (e(H) - \Phi'_H(1-x))\,\mathrm{d}x \right| \\ &\quad \leq \int_{q_2}^{q} 2e(H)^2 x\,\mathrm{d}x \\ &\quad = e(H)^2(q^2 - q_2^2) = e(H)^2 q_1(q+q_2) \leq 2e(H)^2 q q_1. \end{aligned}$$

□

Proposition 4.15. *Suppose H has a vertex of degree at least two, and let $0 < \delta \leq 1/v(H)$. Assume that every graphon V with $\Delta_V \leq \delta$ satisfies*

$$t(H, 1-V) \geq \Phi_H(1 - t(K_2, V)).$$

Then $t(H, 1-U) \geq \Phi_H(1-q)$ whenever

$$q = t(K_2, U) < \min\left\{ \frac{\delta^2}{2}, \frac{\delta}{32e(H)^2} \right\}. \tag{4.34}$$

If $16e(H)^2\delta \leq 1$, it suffices that $q < \delta^2/2$.

Proof. Let $A = \{x \in [0,1] : d_U(x) > \delta\}$ and let $s = |A|$ be its size. Decompose U as

$$U_1(x,y) = U(x,y)\mathbf{1}_{\{x \in A \text{ or } y \in A\}}, \qquad U_2 = U - U_1,$$

and write $q_i = t(K_2, U_i)$. Then $q_1 + q_2 = q$ and

$$s \leq \frac{q}{\delta} \leq \frac{\delta}{2} \leq \frac{1}{2v(H)}.$$

Also $\Delta_{U_2} \leq \delta$, so $t(H, 1-U_2) \geq \Phi_H(1-q_2)$ by hypothesis. For $x \in A$, the average of $U(x,y)$ over A^c satisfies

$$\frac{1}{1-s}\int_{A^c} U(x,y)\,\mathrm{d}y \geq \frac{d_U(x) - s}{1-s} \geq d_U(x) - s \geq \frac{d_U(x)}{2}. \tag{4.35}$$

Moreover,

$$q_1 = 2\int_A d_U(x)\,\mathrm{d}x - \int_{A\times A} U(x,y)\,\mathrm{d}x\,\mathrm{d}y \le 2\int_A d_U(x)\,\mathrm{d}x. \tag{4.36}$$

For a binomial random variable N with parameters $k \in \mathbb{Z}_{\ge 0}$ and $h \in [0,1]$, we have

$$\mathbb{E}(N-1)_+ = (1-h)^k - 1 + kh \ge (k-1)_+ h^2. \tag{4.37}$$

The expression is zero for $k = 0, 1$; for $k \ge 2$,

$$(1-h)^k - 1 + kh = h\sum_{j=1}^{k-1}\bigl(1-(1-h)^j\bigr) \ge h\sum_{j=1}^{k-1} h = (k-1)h^2.$$

Let N be the number of edges selected according to U_1 as in (4.32). For each $v \in V(H)$, define the event

$$E_v = \{Z_v \in A,\ Z_w \in A^c \text{ for every } w \ne v\}.$$

The events E_v are disjoint, and $(N-1)_+ \ge 0$, so

$$D_H(U_1) = \mathbb{E}(N-1)_+ \ge \sum_{v\in V(H)} \mathbb{E}\bigl[(N-1)_+ \mathbf{1}_{E_v}\bigr].$$

On E_v, only edges incident to v can be selected. Conditional on $Z_v = x \in A$ and all other positions lying in A^c, N is binomial with parameters $d_H(v)$ and $(1-s)^{-1}\int_{A^c} U(x,y)\,\mathrm{d}y$. Since the other positions lie in A^c with probability $(1-s)^{v(H)-1}$, (4.37) and (4.35) give

$$\begin{aligned}
D_H(U_1) &\ge (1-s)^{v(H)-1} \sum_{v\in V(H)} (d_H(v)-1)_+ \int_A \left(\frac{1}{1-s}\int_{A^c} U(x,y)\,\mathrm{d}y\right)^2 \mathrm{d}x \\
&\ge \frac{(1-s)^{v(H)-1}}{4}\left(\sum_{v\in V(H)} (d_H(v)-1)_+\right)\int_A d_U(x)^2\,\mathrm{d}x \\
&\ge \frac{1}{8}\int_A d_U(x)^2\,\mathrm{d}x \ge \frac{\delta}{8}\int_A d_U(x)\,\mathrm{d}x \ge \frac{\delta}{16} q_1.
\end{aligned}$$

Here the degree assumption gives $\sum_v (d_H(v)-1)_+ \ge 1$, and $(1-s)^{v(H)-1} \ge 1-(v(H)-1)s \ge 1/2$.

Since $q \le \delta/(32e(H)^2) \le 1/(2e(H))$, superadditivity and Lemma 4.14 give

$$\begin{aligned}
t(H, 1-U) &\ge t(H, 1-U_2) + D_H(U_1) - e(H)q_1 \\
&\ge \Phi_H(1-q_2) + D_H(U_1) - e(H)q_1 \\
&\ge \Phi_H(1-q) + D_H(U_1) - 2e(H)^2 q q_1 \\
&\ge \Phi_H(1-q) + q_1\left(\frac{\delta}{16} - 2e(H)^2 q\right) \\
&\ge \Phi_H(1-q).
\end{aligned}$$

If $16e(H)^2\delta \le 1$, then $\delta^2/2 \le \delta/(32e(H)^2)$, so $q < \delta^2/2$ suffices. □

4.7 The obstruction for even girth

Lemma 4.16. *Suppose H has girth g. If $A \subseteq E(H)$ and H_A contains a cycle, then $r(A) \ge g-1$. Equality holds if and only if deleting the isolated vertices from H_A leaves a single g-cycle.*

Proof. A component of H_A containing a cycle has at least g vertices. A spanning forest of H_A therefore has at least $g-1$ edges, so $r(A) \ge g-1$. If $r(A) = g-1$, this component has exactly g vertices and all other vertices are isolated. Its cycle uses all g vertices, and an additional edge would be a chord creating a shorter cycle. The converse is immediate. □

Proposition 4.17. *If H has girth g, then*

$$\Phi_H(1-q) - (1-q)^{e(H)} = (-1)^g c_g(H) q^{g-1} + O_H(q^g) \quad \text{as } q \downarrow 0. \tag{4.38}$$

If g is even, the constant graphon $W \equiv 1 - 1/k$ violates (1.4) *for every sufficiently large integer k.*

Proof. Subtracting the binomial expansion of $(1-q)^{e(H)}$ from (4.2) gives

$$\Phi_H(1-q) - (1-q)^{e(H)} = \sum_{A \subseteq E(H)} (-1)^{|A|} (q^{r(A)} - q^{|A|}).$$

If H_A is a forest, the summand is zero. Each of the $c_g(H)$ shortest cycles contributes $(-1)^g(q^{g-1} - q^g)$. By Lemma 4.16, every remaining edge set A containing a cycle satisfies $r(A) \geq g$, proving (4.38).

For even g, the leading coefficient is positive, so $\Phi_H(1-q) > (1-q)^{e(H)}$ for all sufficiently small $q > 0$. For the constant graphon $W \equiv 1 - q$,

$$t(H, W) = (1-q)^{e(H)} < \Phi_H(1-q).$$

Taking $q = 1/k$ and using (1.3) shows that the constant graphon $W \equiv 1 - 1/k$ violates (1.4) for every sufficiently large integer k. □

4.8 Proof of the characterisation

Proof of Theorem 1.6. (iii)⇒(ii). Suppose the girth of H is odd, and set $\delta = (6v(H)^6)^{-1}$. By Proposition 4.12, every graphon V with $\Delta_V \leq \delta$ satisfies $t(H, 1-V) \geq \Phi_H(1 - t(K_2, V))$. The graph H has a vertex of degree at least two, $\delta \leq 1/v(H)$, and

$$16e(H)^2\delta \leq 16 \left(\frac{v(H)^2}{2}\right)^2 \frac{1}{6v(H)^6} = \frac{2}{3v(H)^2} \leq 1.$$

Thus Proposition 4.15 gives $t(H, 1-U) \geq \Phi_H(1-q)$ whenever

$$q < \frac{\delta^2}{2} = \frac{1}{72v(H)^{12}}.$$

Taking $U = 1 - W$ proves (ii) with the threshold (1.7).

(ii)⇒(i). For every sufficiently large integer k and every graphon W of edge density $1 - 1/k$,

$$t(H, W) \geq \Phi_H(1 - 1/k) = t(H, T_k),$$

where the equality is (1.3). With the threshold in (1.7), it suffices that $k > 72v(H)^{12}$.

(i)⇒(iii). Since H is nonbipartite, it contains a cycle. If its girth were even, Proposition 4.17 would contradict (i). Hence its girth is odd. □

5 Graphs on at most six vertices

The classification in Theorem 1.7 is obtained by enumerating the 117 graphs and proving the bound or constructing a counterexample for each. The repository listed in Appendix A contains the complete catalogue, with a formal proof for every entry.

For the positive cases, the general results of Sections 2 and 3 are supplemented by estimates for individual graphs. These estimates express homomorphism densities as integrals with one or more vertex positions fixed, then apply the weighted triangle inequality, moment bounds, and Cauchy–Schwarz. The remaining cases follow from sum-of-squares identities. For the negative cases, finite step graphons give strict counterexamples. The 23 violating graphs are shown in Figure 2.

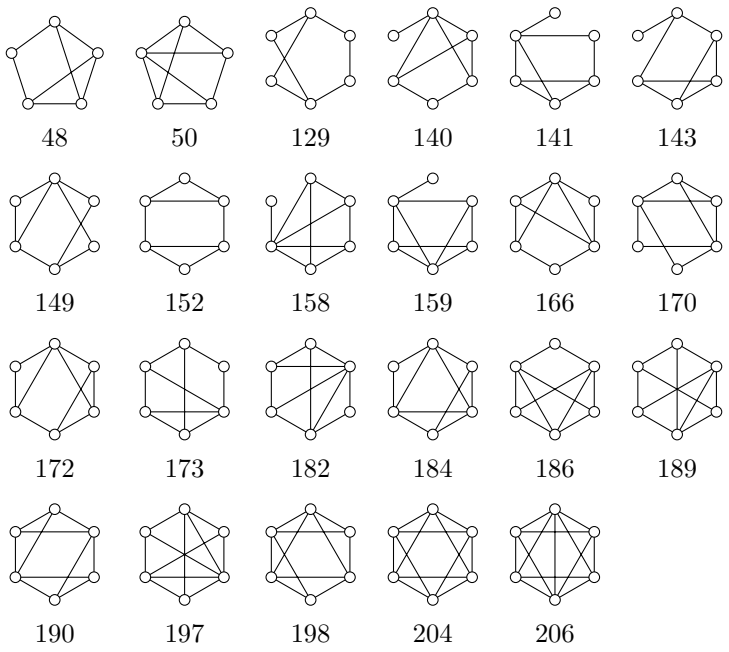


Figure 2: The 23 graphs that violate the Goodman-type lower bound, labelled by their numbers in the Graph Atlas.

5.1 Density ranges for the violating graphs

While the 23 graphs in Figure 2 do not satisfy the Goodman-type lower bound, as Theorem 1.6 proves, they eventually satisfies the lower bound, except for graph 129, whose girth is even. In Table 1, we give partial characterisations of the range on which the bounds hold.

The 94 graphs satisfying (1.2) have optimal T_k for every $k \geq \chi(H)$. The first two rows of Table 1 give six further graphs with this property.

Write H_{173} for graph 173. At edge density 3/4, we have

$$\frac{14056671}{200000000} \leq \inf_{\substack{W \text{ a graphon} \\ t(K_2,W)=3/4}} t(H_{173}, W) \leq \frac{9}{128} = t(H_{173}, T_4). \tag{5.1}$$

The two bounds differ by 5829/200000000. Whether T_4 attains the minimum remains open.

5.2 Constructions of counterexamples

In this subsection, we construct step graphons that violate the Goodman-type lower bound for each of the 23 graphs in Figure 2.

For graphons U and V, their *tensor product* is the graphon on the product probability space $[0,1]^2$ given by

$$(U \otimes V)((x,x'),(y,y')) = U(x,y)V(x',y').$$

Homomorphism densities are multiplicative under this operation:

$$t(H, U \otimes V) = t(H,U)\, t(H,V).$$

For the graphs $129, 149, 170, 173, 186, 189, 197, 198, 204$, the graphon $T_{2k-1} \otimes T_{2k}$, whose edge density is

$$\left(1 - \frac{1}{2k-1}\right)\left(1 - \frac{1}{2k}\right) = 1 - \frac{1}{k},$$

Graph numbers	Lower bound range	Optimality of T_k
152, 166, 172, 182, 184	$p = 2/3$ or $p \geq 3/4$	Optimal for all $k \geq \chi(H) = 3$
206	$p \geq 3/4$	Optimal for all $k \geq \chi(H) = 4$
48, 50, 140, 141, 143, 158, 159, 197, 198	$p \geq 3/4$	Suboptimal at $k = 3$; optimal for $k \geq 4$
186	$p = 3/4$ or $p \geq 4/5$	Suboptimal at $k = 3$; optimal for $k \geq 4$
190	$p \geq 4/5$	Suboptimal at $k = 3, 4$; optimal for $k \geq 5$
173	$p \geq 4/5$	Suboptimal at $k = 3$; optimal for $k \geq 5$; $k = 4$ open
149, 170	$p \geq 1 - (72 \cdot 6^{12})^{-1}$	Suboptimal at $k = 3, 4, 5$; optimal for $k \geq 72 \cdot 6^{12}$
189, 204	$p \geq 1 - (72 \cdot 6^{12})^{-1}$	Suboptimal at $k = 3, 4$; optimal for $k \geq 72 \cdot 6^{12}$
129	None	Suboptimal at every $k \geq 4$

Table 1: Density ranges and Turán optimality for the 23 graphs that violate the Goodman-type lower bound.

satisfies $t(H, T_{2k-1} \otimes T_{2k}) < t(H, T_k)$ for each k listed as suboptimal in Table 1. At $k = 3$, the same construction also gives a counterexample for graph 190.

For graphs 182 and 184, take $W = T_4 \otimes T_4$. Then

$$t(K_2, W) = \frac{9}{16}, \qquad t(H, W) = \frac{81}{65536} < \frac{351}{262144} = \Phi_H\left(\frac{9}{16}\right).$$

For graph 129, the constant graphon $W \equiv 1 - 1/k$ gives

$$t(H, T_k) - t(H, W) = \frac{(k-1)(k^3 - 5k^2 + 6k - 1)}{k^7} > 0 \qquad (k \geq 4).$$

Indeed, substituting $k = u + 4$ turns the cubic factor into $u^3 + 7u^2 + 14u + 7$.

For graphs 166 and 172, partition $[0, 1]$ into four equal parts A_1, A_2, A_3, A_4. Define W to be constant on each $A_i \times A_j$, with value given by the (i, j)-entry of the following matrix:

$$\begin{pmatrix} 0 & \frac{1}{4} & 1 & 1 \\ \frac{1}{4} & 0 & 1 & 1 \\ 1 & 1 & 0 & \frac{1}{4} \\ 1 & 1 & \frac{1}{4} & 0 \end{pmatrix}.$$

Its edge density is $9/16$, and

$$t(H, W) = \begin{cases} 465/262144, & \text{for graph 166,} \\ 3/2048, & \text{for graph 172,} \end{cases} \quad < \frac{477}{262144} = \Phi_H\left(\frac{9}{16}\right).$$

For graph 206, partition $[0, 1]$ into three equal parts A_1, A_2, A_3. Define W to be constant on each $A_i \times A_j$, with value given by the (i, j)-entry of

$$\begin{pmatrix} \frac{1}{4} & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Then

$$t(K_2, W) = \frac{25}{36}, \qquad t(H, W) = \frac{77731841}{48922361856} < \frac{16975}{10077696} = \Phi_H\left(\frac{25}{36}\right).$$

For graph 190 at $k = 4$, partition $[0, 1]$ into four equal parts A_1, A_2, A_3, A_4. For $0 \le \varepsilon \le 1$, define W_ε to be constant on each $A_i \times A_j$, with value given by the (i, j)-entry of the following matrix:

$$\begin{pmatrix} \varepsilon & 1-\varepsilon/3 & 1-\varepsilon/3 & 1-\varepsilon/3 \\ 1-\varepsilon/3 & \varepsilon & 1-\varepsilon/3 & 1-\varepsilon/3 \\ 1-\varepsilon/3 & 1-\varepsilon/3 & \varepsilon & 1-\varepsilon/3 \\ 1-\varepsilon/3 & 1-\varepsilon/3 & 1-\varepsilon/3 & \varepsilon \end{pmatrix}.$$

The edge density remains $3/4$, while $\frac{\mathrm{d}}{\mathrm{d}\varepsilon} t(H, W_\varepsilon)\Big|_{\varepsilon=0} = -1/256$, so sufficiently small positive ε decreases the H-density.

For graphs $48, 50, 140, 141, 143, 158$ at $k = 3$, use the following four-part graphon. Partition $[0, 1]$ into a set A_1 of size $\delta = 1/2000$ and sets A_2, A_3, A_4 of size $(1 - \delta)/3$ each. Define W to be constant on each product of two parts, with values given by the following matrix:

$$\begin{pmatrix} 0 & 1 & \frac{23}{50} & \frac{23}{50} \\ 1 & \varepsilon & 1 & 1 \\ \frac{23}{50} & 1 & \varepsilon & 1 \\ \frac{23}{50} & 1 & 1 & \varepsilon \end{pmatrix}.$$

The choice $\varepsilon = 8046/99900025$ makes the edge density exactly $2/3$. For each of the six graphs, $t(H, W) < t(H, T_3)$; the gaps $t(H, T_3) - t(H, W)$ for graphs $48, 50, 140, 141, 143, 158$ are approximately 4.10×10^{-7}, 2.24×10^{-8}, 3.30×10^{-7}, 1.16×10^{-7}, 8.00×10^{-8}, and 1.08×10^{-7}, respectively.

For graph 159, a three-part graphon gives a counterexample at $k = 3$. Partition $[0, 1]$ into sets A_1, A_2, A_3 of sizes $13/40$, $7/20$, $13/40$, respectively. Define W to be constant on each $A_i \times A_j$, with value given by the (i, j)-entry of the following matrix:

$$\begin{pmatrix} 0 & 1 & \frac{49}{50} \\ 1 & \frac{557}{14700} & 1 \\ \frac{49}{50} & 1 & 0 \end{pmatrix}.$$

Its edge density is exactly $2/3$, and $t(H, W) < t(H, T_3) = 4/243$, with $t(H, T_3) - t(H, W) \approx 1.744 \times 10^{-6}$.

For graph 152, let $0 < \varepsilon < 1/2$ and partition $[0, 1]$ into sets A_1, A_2, B, S_1, S_2 of sizes

$$\left(\frac{1}{2} - \varepsilon\right)\frac{624}{625}, \quad \left(\frac{1}{2} - \varepsilon\right)\frac{1}{625}, \quad \frac{1}{2}, \quad \frac{\varepsilon}{10}, \quad \frac{9\varepsilon}{10}.$$

Define W_ε to be constant on each product of two parts, with values given by the following matrix, whose rows and columns are ordered as A_1, A_2, B, S_1, S_2:

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & \frac{1}{16} & 1 \\ 1 & 0 & \frac{1}{16} & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix}.$$

Let $p_\varepsilon = t(K_2, W_\varepsilon)$. Summing over the partition classes gives

$$p_\varepsilon = \frac{1}{2} + \frac{753}{100000}\varepsilon - \frac{633}{3125}\varepsilon^2,$$
$$t(H, W_\varepsilon) = \frac{2224881}{80000000000}\varepsilon^2(1 - 2\varepsilon)^2.$$

For this graph, the Goodman-type bound (1.2) would require

$$t(H, W_\varepsilon) \geq p_\varepsilon(2p_\varepsilon - 1)^2(3p_\varepsilon^2 - 3p_\varepsilon + 1) \qquad \text{whenever } p_\varepsilon > \frac{1}{2}.$$

The formula for p_ε shows that $p_\varepsilon > 1/2$ for all sufficiently small positive ε. Moreover,

$$\begin{aligned}\lim_{\varepsilon \downarrow 0} \frac{p_\varepsilon(2p_\varepsilon - 1)^2(3p_\varepsilon^2 - 3p_\varepsilon + 1)}{\varepsilon^2} &= \frac{1}{2}\left(\frac{753}{50000}\right)^2 \frac{1}{4} \\ &= \frac{2268036}{80000000000} \\ &> \frac{2224881}{80000000000} = \lim_{\varepsilon \downarrow 0} \frac{t(H, W_\varepsilon)}{\varepsilon^2}.\end{aligned}$$

Hence, for all sufficiently small positive ε,

$$t(H, W_\varepsilon) < p_\varepsilon(2p_\varepsilon - 1)^2(3p_\varepsilon^2 - 3p_\varepsilon + 1),$$

so W_ε is a counterexample to (1.2).

6 Open problems

Theorems 1.1 and 1.3 establish the bound for two classes of chordal graphs. The following construction shows that attaching leaves to a clique can produce a chordal graph that violates the bound.

Proposition 6.1. *Let L_{20} be obtained from a triangle by attaching twenty pendant leaves to each of its three vertices. Then L_{20} is a connected chordal graph with 63 vertices and 63 edges that does not satisfy the Goodman-type lower bound.*

Proof. The polynomial $\Phi_{L_{20}}$ is

$$\Phi_{L_{20}}(p) = p^{61}(2p - 1).$$

Partition $[0, 1]$ into sets A, B of sizes $2/5, 3/5$, and define a graphon W by

$$W|_{A\times A} = 0, \qquad W|_{A\times B} = W|_{B\times A} = 1, \qquad W|_{B\times B} = \frac{1}{9}.$$

Its edge density is $p = 13/25 > 1/2$, and a direct computation gives

$$t(L_{20}, W) \approx 1.01054 \times 10^{-19} < 1.89786 \times 10^{-19} \approx \Phi_{L_{20}}(13/25).$$

Thus W violates the Goodman-type lower bound for L_{20}. □

Problem 1 (Chordal classification)**.** Characterise the connected nonbipartite chordal graphs that satisfy the Goodman-type lower bound. Can a necessary and sufficient condition be expressed in terms of their maximal cliques and the way those cliques intersect?

The graphon in the proof of Proposition 6.1 has edge density $13/25$, which is not a Turán density, so it does not rule out the optimality of T_k for L_{20}. We ask whether T_k is optimal for every graph in a larger class, which contains every chordal graph and every odd cycle.

Problem 2 (Graphs with no induced even cycle)**.** Let H be a nonbipartite graph with no induced even cycle. Is T_k optimal for every integer $k \geq \chi(H)$? Equivalently, does

$$t(H, W) \geq \frac{P_H(k)}{k^{v(H)}}$$

hold for every integer $k \geq \chi(H)$ and every graphon W with $t(K_2, W) = 1 - 1/k$?

A shortest cycle of such a graph is induced and hence odd, so Theorem 1.6 gives the optimality of T_k for all sufficiently large k.

The graphs of Theorem 1.7 also support this question. Each of the 23 graphs in Figure 2 contains an induced C_4. Hence all 75 of the 117 graphs with no induced even cycle satisfy the Goodman-type lower bound.

We finally ask for the optimal exponent of $v(H)$ in the density threshold of Theorem 1.6. The following construction shows that every exponent below 4 is insufficient.

Proposition 6.2 (Quartic obstruction)**.** *For every sufficiently large odd integer n, there is an n-vertex graph H of girth 3 and chromatic number 3 such that the constant graphon $W \equiv p$, where*

$$p = 1 - \frac{128}{(n-1)^4},$$

satisfies $t(H, W) < \Phi_H(p)$.

Proof. Let $r = (n-1)/2$ and let H be the graph obtained from $K_{r,r}$ by adding one vertex adjacent to both ends of a chosen edge. Denote this graph by J_r. The new vertex completes a triangle and can receive a third colour in a 2-colouring of $K_{r,r}$. Thus J_r has girth 3 and chromatic number 3. Moreover, $v(J_r) = 2r+1$, $e(J_r) = r^2+2$, and $P_{J_r}(x) = (x-2)P_{K_{r,r}}(x)$. Let $\alpha_1, \dots, \alpha_{2r}$ be the roots of $P_{K_{r,r}}$. Sokal's bound [17] states that every root of the chromatic polynomial of a graph has modulus at most eight times its maximum degree. Since $K_{r,r}$ has maximum degree r, we have $|\alpha_i| \le 8r$ for every i. Since $P_{K_{r,r}}$ is monic of degree $2r$,

$$\Phi_{K_{r,r}}(1-q) = q^{2r} P_{K_{r,r}}(1/q) = \prod_{i=1}^{2r} (1 - \alpha_i q).$$

For $0 < q \le 1/(16r)$, we have $|\alpha_i q| \le 1/2$, so the convergent series $\log(1-z) = -\sum_{j\ge1} z^j/j$ gives

$$\begin{aligned}
\log \frac{\Phi_{K_{r,r}}(1-q)}{(1-q)^{r^2}} &= \sum_{i=1}^{2r} \log(1-\alpha_i q) - r^2 \log(1-q) \\
&= \sum_{j\ge1} \frac{r^2 - \sum_{i=1}^{2r} \alpha_i^j}{j} q^j \\
&= \left(r^2 - \sum_{i=1}^{2r} \alpha_i\right) q + \frac{1}{2}\left(r^2 - \sum_{i=1}^{2r} \alpha_i^2\right) q^2 \\
&\quad + \frac{1}{3}\left(r^2 - \sum_{i=1}^{2r} \alpha_i^3\right) q^3 + O(r^5 q^4).
\end{aligned}$$

Since $K_{r,r}$ has r^2 edges and no triangles, (4.2) and Vieta's formulas give

$$\sum_{i=1}^{2r} \alpha_i = \sum_{\substack{A \subseteq E(K_{r,r}) \\ |A|=1}} 1 = r^2, \qquad \sum_{1\le i<j\le 2r} \alpha_i \alpha_j = \sum_{\substack{A \subseteq E(K_{r,r}) \\ |A|=2}} 1 = \binom{r^2}{2}.$$

Hence

$$\sum_{i=1}^{2r} \alpha_i^2 = \left(\sum_{i=1}^{2r} \alpha_i\right)^2 - 2 \sum_{1\le i<j\le 2r} \alpha_i \alpha_j = r^4 - 2\binom{r^2}{2} = r^2.$$

Thus the terms with $j = 1$ and $j = 2$ are zero.

For the cubic term, the edge subsets contributing to the coefficient of q^3 in (4.2) are precisely the three-edge forests and the four-cycles, with signs -1 and $+1$, respectively. Every three-edge subset of $E(K_{r,r})$ is a forest. Each four-cycle is obtained by choosing two vertices from each part, giving $\binom{r}{2}^2$ choices. Vieta's formulas therefore give

$$\sum_{1\le i<j<k\le 2r} \alpha_i\alpha_j\alpha_k = \sum_{\substack{A\subseteq E(K_{r,r})\\ |A|=3}} 1 - \binom{r}{2}^2 = \binom{r^2}{3} - \binom{r}{2}^2.$$

It follows that

$$\begin{aligned}\sum_{i=1}^{2r}\alpha_i^3 &= \left(\sum_i \alpha_i\right)^3 - 3\left(\sum_i \alpha_i\right)\left(\sum_{i<j}\alpha_i\alpha_j\right) + 3\sum_{i<j<k}\alpha_i\alpha_j\alpha_k \\ &= r^6 - 3r^2\binom{r^2}{2} + 3\left(\binom{r^2}{3} - \binom{r}{2}^2\right) \\ &= r^2 - 3\binom{r}{2}^2.\end{aligned}$$

Thus the coefficient of q^3 in the logarithmic expansion is

$$\frac{r^2 - \sum_{i=1}^{2r}\alpha_i^3}{3} = \binom{r}{2}^2.$$

For the remaining terms, $|\sum_i \alpha_i^j| \le 2r(8r)^j$ gives

$$\begin{aligned}\left|\sum_{j\ge 4}\frac{r^2 - \sum_{i=1}^{2r}\alpha_i^j}{j}q^j\right| &\le \frac{r^2q^4}{4(1-q)} + \frac{r(8rq)^4}{2(1-8rq)} \\ &\le \frac{1}{2}r^2q^4 + r(8rq)^4 = O(r^5q^4),\end{aligned}$$

where the last inequality follows from $q \le 1/2$ and $8rq \le 1/2$. Consequently,

$$\begin{aligned}\log\frac{\Phi_{J_r}(1-q)}{(1-q)^{r^2+2}} &= \log\frac{1-2q}{(1-q)^2} + \log\frac{\Phi_{K_{r,r}}(1-q)}{(1-q)^{r^2}} \\ &= -q^2 + \left(\binom{r}{2}^2 - 2\right)q^3 + O(r^5q^4).\end{aligned}$$

Taking $q = 8/r^4$ gives

$$\log\frac{\Phi_{J_r}(p)}{p^{r^2+2}} = \frac{64}{r^8} + O(r^{-9}) > 0$$

for all sufficiently large r. Thus $t(J_r, W) = p^{r^2+2} < \Phi_{J_r}(p)$. □

Together, Theorem 1.6 and Proposition 6.2 show that the optimal uniform exponent lies between 4 and 12.

Problem 3 (Optimal exponent in the density threshold)**.** Determine the infimum of the exponents γ for which there is a constant $C > 0$ such that

$$t(H, W) \ge \Phi_H(p) \qquad \text{whenever} \qquad p = t(K_2, W) > 1 - \frac{1}{Cn^\gamma},$$

for every integer $n \ge 3$, every graph H of odd girth with at most n vertices, and every graphon W. In particular, does such a bound hold with $\gamma = 4$?

A Formal verification

The accompanying Lean 4 development, using Mathlib, is available at

`https://github.com/mekty2012/GoodmanTypeBound`.

It verifies Theorems 1.1, 1.3, 1.4, 1.6 and 1.7, Lemma 1.2, Corollary 1.5, and Proposition 6.1. The module `Taeyoung/Paper.lean` records the correspondence between these statements and their Lean declarations.

The odd-cycle theorem on the full range $p > 1/2$ is supplied by the formalisation of Im, Kim, and Yang accompanying [6]. It is imported as the Lake dependency `OddCycleBound`, pinned to commit `dbaea5a`. The proof of Theorem 1.6 follows the forest expansion and quantitative estimates of Section 4. For Theorem 1.7, the development contains a proof for each of the 117 graphs, including the sum-of-squares certificates used in the positive cases and exact counterexamples for the negative cases.

The density ranges and Turán-optimality results in Table 1, together with the counterexamples in Section 5.2, are also formalised in Lean. The bounds for $p \geq 3/4$ and $p \geq 4/5$, and at $p = 2/3$ and $p = 3/4$, follow from rational sum-of-squares certificates on six-vertex samples. The ranges $p \geq 1-(72\cdot 6^{12})^{-1}$ follow from Theorem 1.6 and continuity at the endpoint. Among these results, only the lower bound in (5.1) has not been formalised in Lean. It follows from a certificate on seven-vertex samples checked by an exact verifier in the repository.

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